NEKHOROSHEV STABILITY FOR QUASI-CONVEX HAMILTONIANS: BLUEPRINT (TRACK N)

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Abstract. A proof plan, step by step, for the three frozen statements of track N of the KAM formalization: the exponential stability estimate for strictly quasi-convex nearly integrable Hamiltonians with a complex Hessian bound (N-HEAD-C, after Pöschel 1993, Theorem 1), the exact integrable dynamics for zero perturbation (N-ZERO) and the nonsingular isoenergetic derivative (N-QC). Every result is labelled and cites its pinned source. This version contains Part 0 only; the plan of the remaining parts is in `PLAN.md`.

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1. Conventions, definitions and the frozen statements

This part fixes notation and restates, in mathematical form, the three frozen theorems of track N (owner approval D20, 27 October 2026) and the eleven frozen definitions of `KAM/Frozen/Nekhoroshev/Defs/` together with the five track-K definitions they use. The Lean declarations are authoritative; the statements below are transcriptions of them, with the Lean names recorded under each. N-HEAD with real bounds only is *not* frozen (source gap F-N1, D17/D18/D20); the frozen headline is N-HEAD-C. The additions of D22 (N-PARTIAL-CONVEX, N-AVERAGING) are not yet frozen and are not part of this part. Results of track K's blueprint are cited as [K:`bp:p1:...`].

1.1. Conventions.

- $n \in \mathbb{N}$ is the number of degrees of freedom; N-HEAD-C assumes $n \geq 3$ (D17), N-ZERO and N-QC are stated for every n. A phase point is $z = (\theta, I)$ with angles θ and actions I, in this order.
- Ambient spaces are those of the Lean text: $\mathbb{R}^n = (\texttt{Fin}\ n \to \mathbb{R})$, $\mathbb{C}^n = (\texttt{Fin}\ n \to \mathbb{C})$, with the max norms $|v| = \max_i |v_i|$ on both. The embedding $\mathbb{R}^n \subset \mathbb{C}^n$ is componentwise. The norm $\|\cdot\|$ in every frozen statement is the max norm; on the phase space $\mathbb{R}^n \times \mathbb{R}^n$ it is $\|(\theta, I)\| = \max(|\theta|, |I|)$ (Lean product norm). The only Euclidean structure is the dot product $\sum_i \omega_i v_i$ used for orthogonality in quasi-convexity; Euclidean norms used later for comparison with the sources come with explicit inequalities ($|v| \leq \|v\|_2 \leq \sqrt{n}|v|$). For $n = 0$ all these spaces are $\{0\}$ and the norm is 0.
- Operator norms of linear and multilinear maps are the induced norms of these max norms. In particular $\|D^2 h(z)\|$ for complex h is the norm of $D^2 h(z) \in \mathcal{L}(\mathbb{C}^n, \mathcal{L}(\mathbb{C}^n, \mathbb{C}))$ (Lean `fderiv` applied twice), i.e. $\sup_{|u|,|v|\leq 1} |D^2 h(z)[u, v]|$.
- Angles are *not* reduced modulo $\mathbb{Z}^n$ in the statements: $\theta \in \mathbb{R}^n$ or $\theta \in T_s \subset \mathbb{C}^n$, and 1-periodicity is a hypothesis on f (Definition 1.12). So the torus is $\mathbb{R}^n/\mathbb{Z}^n$ and the Fourier characters are $e^{2\pi i k \cdot \theta}$ as in track K ([K:`bp:p1:decay`], [K:`bp:p1:summation`]). Pöschel's torus is $\mathbb{R}^n/2\pi\mathbb{Z}^n$ and his norms are Euclidean; the conversion is a separate step of the proof (Part 1 of `PLAN.md`).
- The complex strip is $T_s = \{\theta \in \mathbb{C}^n : |\operatorname{Im} \theta_i| < s\ \forall i\}$ (Lean `KAM.complexStrip`), as in track K.

1.2. Track-K definitions used.

Definition 1.1 (Complex action neighbourhood)**.** For $B \subset \mathbb{R}^n$ and $r \in \mathbb{R}$, $B_r = \{I \in \mathbb{C}^n : \exists J \in B,\ |I - J| < r\}$ (max norm, J viewed in $\mathbb{C}^n$). Thus $B_r = \emptyset$ for $r \leq 0$, and for B open B_r is open.
Frozen Lean declaration: `KAM.actionNbhd`
Source: BM11 p. 4 (D_s); track K, [K:`bp:p0:domains`].

Definition 1.2 (Frequency map)**.** For $h : \mathbb{C}^n \to \mathbb{C}$ and $I \in \mathbb{R}^n$, $\omega_h(I) = \nabla h(I) \in \mathbb{R}^n$ is the vector with $\omega_h(I)_i = D\big(J \mapsto \operatorname{Re} h(J)\big)(I)[e_i]$, the real partial derivatives of $\operatorname{Re} h$ on $\mathbb{R}^n$ at I. It is 0 if $\operatorname{Re} h|_{\mathbb{R}^n}$ is not differentiable at I (Lean totality, Remark 1.18).
Frozen Lean declaration: `KAM.frequencyMap`
Source: BM11 p. 1; Pos93 p. 190 ($\omega(I) = \partial_I h(I)$).

Definition 1.3 (Hamiltonian vector field)**.** For $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ and $z = (\theta, I)$, $X_H(z) = \big((\partial_{I_i} H(z))_i, (-\partial_{\theta_i} H(z))_i\big)$, where $\partial_{I_i} H(z) = DH(z)[(0, e_i)]$ and $\partial_{\theta_i} H(z) = DH(z)[(e_i, 0)]$ are the Fréchet derivatives in the Lean sense (0 if H is not differentiable at z).
Frozen Lean declaration: `KAM.hamiltonianVectorField`
Source: BM11 p. 1; track K, [K:`bp:p0:hamiltonian`].

Definition 1.4 (Real Hamiltonian)**.** For $h : \mathbb{C}^n \to \mathbb{C}$, $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ and real $z = (\theta, I)$, $H_{h,f}(z) = \operatorname{Re}\big(h(I) + f(\theta, I)\big)$, with θ, I viewed in $\mathbb{C}^n$.
Frozen Lean declaration: `KAM.realHamiltonian`
Source: BM11 p. 5 ($H = h + f$).

Remark 1.5 (Strip and neighbourhood notation). `KAM.complexStrip` $n\, s = T_s$ as above; for $n > 0$ it is empty when $s \le 0$, and for $n = 0$ it is the whole (one-point) space $\mathbb{C}^0$ for every s. No hypothesis $n > 0$ is added to N-ZERO or N-QC.
Frozen Lean declaration: `KAM.complexStrip`

1.3. **Frozen track-N definitions.**

Definition 1.6 (Action ball)**.** $B(0, R) = \{I \in \mathbb{R}^n : |I| < R\}$, the open ball of the max norm; written `actionBall` $n\, R$.
Frozen Lean declaration: `KAM.Nekhoroshev.actionBall`
Source: BM11 p. 4.

Definition 1.7 (Complex domain)**.** $D_s(R) = T_s \times B(0, R)_s \subset \mathbb{C}^n \times \mathbb{C}^n$; `complexDomain` $n\, R\, s$. It is the period-one lift of BM11's D_s, with a single width s for angles and actions. It is open.
Frozen Lean declaration: `KAM.Nekhoroshev.complexDomain`
Source: BM11 p. 4 (D_s); ours for the lift.

Definition 1.8 (Real action domain)**.** $\mathcal{B}_s(R) = \{I \in \mathbb{R}^n : I \in B(0, R)_s\}$; `realActionDomain` $n\, R\, s$.
Frozen Lean declaration: `KAM.Nekhoroshev.realActionDomain`
Source: BM11 p. 4 (B_s).

Remark 1.9 (Computing $\mathcal{B}_s(R)$). Since $|I - J|$ for real I, J is the real max norm, $\mathcal{B}_s(R) = \{I \in \mathbb{R}^n : \exists J,\ |J| < R,\ |I - J| < s\} = B(0, R + s)$ for $R, s > 0$. This identity is ours and is proved from the definitions (it is not stated in the source); it shows that $B(0, R) \subset \mathcal{B}_s(R)$ and that $\mathcal{B}_s(R)$ is open.

Definition 1.10 (Real part of the integrable part)**.** For $h : \mathbb{C}^n \to \mathbb{C}$ and $I \in \mathbb{R}^n$, $h_{\mathbb{R}}(I) = \operatorname{Re} h(I)$; `realPart` h.
Frozen Lean declaration: `KAM.Nekhoroshev.realPart`
Source: BM11 p. 4 ($h : B_s \to \mathbb{R}$).

Definition 1.11 (Analytic integrable part, $h \in A_s(D)$)**.** `IsAnalyticIntegrable` $R\, s\, h$ holds if

(i) h is complex differentiable on $B(0, R)_s$;
(ii) h is continuous on $\overline{B(0, R)_s}$ (closure in $\mathbb{C}^n$);
(iii) there is $A \ge 0$ with $|h(I)| \le A$ for all $I \in B(0, R)_s$;
(iv) $\operatorname{Im} h(I) = 0$ for all $I \in \mathcal{B}_s(R)$;
(v) $h_{\mathbb{R}}$ is of class C^3 on $\mathcal{B}_s(R)$ (real Fréchet sense; the set is open).

No numerical bound on h is fixed. Clause (v) is redundant by holomorphy ([K:`bp:p1:holomorphic-regularity`], [K:`bp:p1:real-restriction`]) and records the regularity that guards every real derivative.

Frozen Lean declaration: `KAM.Nekhoroshev.IsAnalyticIntegrable`
Source: BM11 p. 4 ($A_s(D)$, integrable case, with continuity on the closure).

Definition 1.12 (Analytic perturbation of size ε)**.** `IsAnalyticPerturbation` $R\,s\,\varepsilon\,f$ holds if $\varepsilon \geq 0$ and

(i) f is complex differentiable on $D_s(R)$ and continuous on $\overline{D_s(R)}$;
(ii) $f(\theta + k, I) = f(\theta, I)$ for all $(\theta, I) \in D_s(R)$ and $k \in \mathbb{Z}^n$;
(iii) $\operatorname{Im} f(\theta, I) = 0$ for all $\theta \in \mathbb{R}^n$ and $I \in \mathcal{B}_s(R)$;
(iv) $|f(z)| \leq \varepsilon$ for all $z \in D_s(R)$ (complex absolute value).

The size is measured by $\leq \varepsilon$, not the source's strict bound; the source is applied with 2ε. For $\varepsilon = 0$ this forces $f = 0$ on $D_s(R)$.
Frozen Lean declaration: `KAM.Nekhoroshev.IsAnalyticPerturbation`
Source: BM11 pp. 4–5 ($f \in A_s(D)$, $|f|_s < \varepsilon$); deviation: $\leq \varepsilon$ (ruling Q2, D18).

Definition 1.13 (Derivative bounds $B(M)$)**.** For $U \subset \mathbb{R}^n$, $h : \mathbb{C}^n \to \mathbb{C}$ and $M \in \mathbb{R}$, `HasDerivativeBounds` $U\,h\,M$ holds if $M > 0$, U is open, $h_{\mathbb{R}} \in C^3(U)$, and for all $I \in U$, all $k \in \{1, 2, 3\}$ and all $(i_1, \ldots, i_k) \in \{1, \ldots, n\}^k$,

$$\left|D^k h_{\mathbb{R}}(I)[e_{i_1}, \ldots, e_{i_k}]\right| \leq M.$$

These are all ordered coordinate partials of orders one to three; for C^3 functions they coincide with the multi-index partials $\partial^k h$ of the source (symmetry, [K:`bp:p1:ext-schwarz`]). The bound is on coordinate partials, not on operator norms.
Frozen Lean declaration: `KAM.Nekhoroshev.HasDerivativeBounds`
Source: BM11 p. 4, $B(M)$.

Definition 1.14 (Strict quasi-convexity $QC(m)$)**.** For $U \subset \mathbb{R}^n$, $h : \mathbb{C}^n \to \mathbb{C}$, $m \in \mathbb{R}$, `IsStrictlyQuasiConvex` $U\,h\,m$ holds if $m > 0$, U is open, $h_{\mathbb{R}} \in C^2(U)$, and for every $I \in U$: $\omega_h(I) \neq 0$ and

$$\sum_i \omega_h(I)_i v_i = 0 \implies m|v|^2 \leq D^2 h_{\mathbb{R}}(I)[v, v] \qquad (v \in \mathbb{R}^n),$$

with $|v|$ the max norm on the left and the Euclidean dot product in the hypothesis.
Frozen Lean declaration: `KAM.Nekhoroshev.IsStrictlyQuasiConvex`
Source: BM11 p. 4, $QC(m)$ (the source's norm on the right is the max norm, orthogonality is Euclidean).

Definition 1.15 (Complex Hessian bound)**.** `HasComplexHessianBound` $R\,s\,h\,M_{\mathbb{C}}$ holds if $M_{\mathbb{C}} > 0$, h is C^2 in the complex sense on $B(0, R)_s$ and $\|D^2 h(z)\| \leq M_{\mathbb{C}}$ for all $z \in B(0, R)_s$, the norm being that of $D(Dh)(z) \in \mathcal{L}(\mathbb{C}^n, \mathcal{L}(\mathbb{C}^n, \mathbb{C}))$ for the max norm. Every entry $|\partial_i \partial_j h|$ is then $\leq M_{\mathbb{C}}$, and the Euclidean operator norm of the Hessian matrix is $\leq n M_{\mathbb{C}}$. This is not the spectral radius of a complex symmetric matrix.
Frozen Lean declaration: `KAM.Nekhoroshev.HasComplexHessianBound`
Source: Pos93 p. 190, "The Hessian of the integrable hamiltonian h" (conversion by n recorded in `docs/tracks/N/sources.md`, anchor POS190-HESS).

Definition 1.16 (Solutions on a time set)**.** For $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, $U \subset \mathbb{R}^n$, $J \subset \mathbb{R}$ and $z : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$, `IsSolutionOn` $H\,U\,J\,z$ holds if for every $t \in J$: $z(t)_2 \in U$, H is differentiable at $z(t)$, and z is differentiable at t (two-sided, also at endpoints of J, since z is defined on $\mathbb{R}$) with $z'(t) = X_H(z(t))$.
Frozen Lean declaration: `KAM.Nekhoroshev.IsSolutionOn`
Source: BM11 Theorem 2.1 p. 5 and the maximal-solution argument p. 13.

Definition 1.17 (The integrable orbit)**.** For $h : \mathbb{C}^n \to \mathbb{C}$, $\theta_0, I_0 \in \mathbb{R}^n$ and $t \in \mathbb{R}$: $\gamma_{\theta_0, I_0}(t) = (\theta_0 + t\,\omega_h(I_0),\ I_0)$.
Frozen Lean declaration: `KAM.Nekhoroshev.integrableOrbit`
Source: BM11 p. 1 ($H(\theta, I) = h(I)$).

Remark 1.18 (Lean totality). The frozen definitions and theorems use `fderiv` (which is 0 at a point of non-differentiability), `iteratedFDeriv`, the operator norm, `rpow` ($\varepsilon^{\pm 1/(2n)}$), `Real.exp` and the max norm; there is no matrix inverse, no Bochner integral and no division by a quantity that may vanish. The guards are:

(a) In N-HEAD-C and N-ZERO, ω_h and $D^2 h_{\mathbb{R}}$ are used only on the open set $\mathcal{B}_s(R)$ where $h_{\mathbb{R}}$ is C^3 (Definitions 1.11(v), 1.13, 1.14), so the derivatives are the genuine ones. N-QC uses them on its arbitrary open U, where its hypothesis gives C^2 (item (e)). The definitions `frequencyMap`, `integrableOrbit` themselves are total and keep Lean's default values outside the differentiability locus.
(b) $D^2 h$ for complex h is used only on $B(0,R)_s$, open, where h is C^2 (Definition 1.15).
(c) X_H is used only inside `IsSolutionOn`, which requires differentiability of H at every visited point; in N-ZERO and N-HEAD-C this is a proof obligation: $H_{h,f}$ is real-differentiable on $\mathbb{R}^n \times \mathcal{B}_s(R) \supseteq \mathbb{R}^n \times B(0,R)$ (angles first, actions restricted) because h, f are holomorphic on open sets containing the real points ([K:`bp:p1:hamiltonian-real`]).
(d) $\varepsilon^{-1/(2n)}$ and $\varepsilon^{1/(2n)}$ occur only with $\varepsilon > 0$ in N-HEAD-C, and $2n \neq 0$.
(e) In N-QC the map $\Psi(I', \lambda) = (h_{\mathbb{R}}(I'), \lambda \omega_h(I'))$ is differentiable at (I, λ) because $h_{\mathbb{R}} \in C^2$ on the open set $U \ni I$; `fderiv` is then the true derivative.

1.4. **The frozen theorems.**

Theorem 1.19 (N-HEAD-C: exponential stability, complex Hessian bound)**.** *Let $n \geq 3$ and $R, s, M, m, M_{\mathbb{C}}, \omega_{\min} > 0$. Then there are $C, T, c, \varepsilon_0 > 0$ with*

$$C\, \varepsilon_0^{1/(2n)} \leq R/4$$

such that the following holds for every $h : \mathbb{C}^n \to \mathbb{C}$, $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ and $\varepsilon \in \mathbb{R}$ with

(H1) `IsAnalyticIntegrable` $R\, s\, h$ *(Definition 1.11);*
(H2) `HasComplexHessianBound` $R\, s\, h\, M_{\mathbb{C}}$ *(Definition 1.15);*
(H3) $\omega_{\min} \leq |\omega_h(I)|$ *for every $I \in B(0,R)$ (max norm);*
(H4) `IsStrictlyQuasiConvex` $\mathcal{B}_s(R)\, h\, m$ *(Definition 1.14);*
(H5) `HasDerivativeBounds` $\mathcal{B}_s(R)\, h\, M$ *(Definition 1.13);*
(H6) `IsAnalyticPerturbation` $R\, s\, \varepsilon\, f$ *(Definition 1.12);*
(H7) $0 < \varepsilon \leq \varepsilon_0$.

For every $\theta_0 \in \mathbb{R}^n$ and $I_0 \in B(0, R/2)$, with $\tau := T \exp\bigl(c\, \varepsilon^{-1/(2n)}\bigr)$ and $H = H_{h,f}$, there is $z : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$ with

(a) $z(0) = (\theta_0, I_0)$;
(b) `IsSolutionOn` $H\, B(0,R)\, [-\tau, \tau]\, z$;
(c) $|z(t)_2 - I_0| \leq C \varepsilon^{1/(2n)}$ *for all t with $|t| \leq \tau$;*
(d) (uniqueness) for every $L \in [0, \tau]$ and every $y : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$ with $y(0) = (\theta_0, I_0)$ and `IsSolutionOn` $H\, B(0,R)\, [-L, L]\, y$*: $y(t) = z(t)$ for all $t \in [-L, L]$.*

The order of choice is: n; $R, s, M, m, M_{\mathbb{C}}, \omega_{\min}$; then C, T, c, ε_0, depending only on these seven data; then h, f, ε, then θ_0, I_0; the solution z may depend on all of them. There is no dependence on A in Definition 1.11(iii). In (H4)–(H5) the set is $\mathcal{B}_s(R) = B(0, R+s)$ (Remark 1.9), in (H3) the smaller ball $B(0,R)$.

Frozen Lean declaration: `KAM.Nekhoroshev.quasiConvex_nekhoroshev_complexBound`

Source: Pos93 Theorem 1, pp. 190–191 (with the conversions and anchors of `docs/tracks/N/sources.md`*); exponent $1/(2n)$ as BM11 p. 5 with $\delta = (2n(n-1))^{-1}$; ruling D18 Q2, Q4; deviation D20: the extra inputs $M_{\mathbb{C}}$ and $\omega_{\min}$.*

Source gap: The printed theorem is for a 2π-periodic torus, Euclidean norms and exceptional initial frequencies $\|\omega(I_0)\| \leq m r_0/8$; the conversion to the frozen statement (rescaling $H \mapsto 2\pi H(\varphi/2\pi, I)$, choice of r_0, s_0

from the seven data, exclusion of the exceptional branch by $\omega_{\min}$, existence and uniqueness of the solution) is given only in `sources.md` *and is proved in Part 10 of* `PLAN.md`*.*

Theorem 1.20 (N-ZERO: exact integrable dynamics)**.** *Let $n \in \mathbb{N}$, $R, s > 0$, h with* `IsAnalyticIntegrable` $R\,s\,h$ *and f with* `IsAnalyticPerturbation` $R\,s\,0\,f$*. For every $\theta_0 \in \mathbb{R}^n$ and $I_0 \in B(0, R/2)$, with $\gamma = \gamma_{\theta_0, I_0}$ (Definition 1.17):*

$$\gamma(0) = (\theta_0, I_0), \qquad \texttt{IsSolutionOn}\, H_{h,f}\, B(0,R)\, \mathbb{R}\, \gamma, \qquad \gamma(t)_2 = I_0 \quad (t \in \mathbb{R}).$$

Frozen Lean declaration: `KAM.Nekhoroshev.zero_perturbation`
Source: Ours (supplement to BM11 Theorem 2.1, p. 5, whose strict bound excludes $\varepsilon = 0$); from $H = h(I)$, BM11 p. 1.

Theorem 1.21 (N-QC: nonsingular isoenergetic derivative)**.** *Let $n \in \mathbb{N}$, $U \subset \mathbb{R}^n$, $h : \mathbb{C}^n \to \mathbb{C}$, $m \in \mathbb{R}$ with* `IsStrictlyQuasiConvex` $U\,h\,m$*. For $I \in U$ and $\lambda > 0$, the derivative at (I, λ) of*

$$\Psi : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R} \times \mathbb{R}^n, \qquad \Psi(I', \lambda') = \big(h_{\mathbb{R}}(I'),\ \lambda' \omega_h(I')\big)$$

is a bijective linear map $\mathbb{R}^n \times \mathbb{R} \to \mathbb{R} \times \mathbb{R}^n$. Explicitly it is $D\Psi(I,\lambda)[v,u] = \big(\omega_h(I) \cdot v,\ u\,\omega_h(I) + \lambda D^2 h_{\mathbb{R}}(I) v\big)$. No uniform radius of invertibility is asserted.
Frozen Lean declaration: `KAM.Nekhoroshev.quasiConvex_isoenergetic_derivative`
Source: BM11 Lemma 3.3 and its proof, pp. 9–10 (only the derivative assertion).

2. Interfaces and conversion of hypotheses

This part connects the frozen hypotheses of track N (Definitions of Part 0) to the form in which the proofs use them. It proves: the real action domain and the inclusions of tubes (§2.2); that holomorphy gives a smooth real Hamiltonian with $X_H = X_{h+f}$ (§2.3); the translation of the derivative bounds $B(M)$, the complex Hessian bound and $QC(m)$ into the Euclidean, uniform form of Pöschel's Theorem 1 (§2.4); the change of normalisation 2π-periodic$\leftrightarrow$ 1-periodic and the sup-to-weighted Fourier norm (§2.5); and the dictionary of constants (§2.6). *Convention ruling.* All of Parts 1–10 work in the frozen conventions (period 1, max norms, characters $e^{2\pi i k\cdot\theta}$). Pöschel's normalisation (period 2π, Euclidean norms on actions, weights $e^{|k|_1 s}$) enters only through the labelled conversions Lemmas 2.8, 2.9, 2.7, 2.10; quantities in his normalisation carry a tilde or the subscript P. This overrules the suggestion in `PLAN.md` to work in Pöschel's normalisation. The Euclidean norm is written $|\cdot|_2$, the max norm $|\cdot|$ (or $|\cdot|_\infty$), and $|k|_1 = \sum_i |k_i|$; the phase-space norm is $\max(|\theta|_\infty, |I|_\infty)$, Euclidean structure is used only for the dot product and where stated.

2.1. External facts.

(E1) *Finite-dimensional linear algebra.* An injective linear map between real vector spaces of equal finite dimension is bijective (rank–nullity; Mathlib `LinearMap.injective_iff_surjective`). Exists.

(E2) *Norm comparison and Cauchy–Schwarz.* For $x \in \mathbb{R}^n$ or $\mathbb{C}^n$: $|x|_\infty \le |x|_2 \le \sqrt{n}|x|_\infty$ and $|x|_1 \le \sqrt{n}|x|_2$, $|x|_2 \le |x|_1$; $|x \cdot y| \le |x|_2 |y|_2$ (for the bilinear, not Hermitian, product too). Exists.

(E3) *Restriction of scalars and chain rule.* A $\mathbb{C}$-differentiable (resp. C^k) map is $\mathbb{R}$-differentiable (resp. C^k) with the same derivative; the chain rule for the $\mathbb{R}$-linear inclusion $\iota : \mathbb{R}^N \to \mathbb{C}^N$ and for Re. (Mathlib `DifferentiableAt.restrictScalars`, `ContDiffOn.restrict_scalars`, `HasFDerivAt.comp`.) Exists.

(E4) *Product rule* for $(x, \lambda) \mapsto \lambda\, v(x)$ with v differentiable, and differentiability of a vector function coordinatewise. Exists.

Track-K results used: [K:`bp:p1:holomorphic-regularity`], [K:`bp:p1:real-restriction`], [K:`bp:p1:hamiltonian-real`], [K:`bp:p1:ext-schwarz`], [K:`bp:p1:decay`], [K:`bp:p1:summation`], [K:`bp:p1:exp-sums`].

2.2. **Domains and norms.**

Lemma 2.1 (Norm and lattice comparisons). *(a) For $x \in \mathbb{R}^n$ (or $\mathbb{C}^n$): $|x|_\infty \le |x|_2 \le \sqrt{n}\,|x|_\infty$, hence $|x|_\infty^2 \ge |x|_2^2/n$ and $|x|_\infty \le |x|_2$. (b) For $k \in \mathbb{Z}^n$: $|k|_2 \le |k|_1 \le \sqrt{n}|k|_2$. Consequently $\{|k|_1 \le K\} \subset \{|k|_2 \le K\}$ and $\{|k|_2 \le K\} \subset \{|k|_1 \le \sqrt{n}K\}$; so a geometric cutoff K_g in the Euclidean norm and an analytic cutoff K_a in $|\cdot|_1$ are compatible when $\sqrt{n}\,K_g \le K_a$. (c) The bilinear bound: if $|Q_{ij}| \le M_0$ for a matrix Q then $|x^\top Q y| \le n M_0 |x|_2 |y|_2$ for $x, y \in \mathbb{C}^n$.*
Source: Standard ((E2)); the lattice inclusions correct the direction stated in `PLAN.md` (review N06).

Proof. **Step S2.1: Vectors.** (a),(b) are (E2): $|x|_2^2 = \sum x_i^2 \le n \max x_i^2$, $|x|_1 \le \sqrt{n}|x|_2$ is Cauchy–Schwarz with $(1, \dots, 1)$.

Step S2.2: Matrices. $|x^\top Q y| \le M_0 |x|_1 |y|_1 \le M_0\, n |x|_2 |y|_2$ by (b). □

Lemma 2.2 (Domains). *Let $R, s > 0$, $n \ge 0$.*

- *(a) $\mathcal{B}_s(R) = B(0, R+s)$; in particular it is open and $B(0,R) \subset \mathcal{B}_s(R)$ (Remark 1.9); $W_s := \mathbb{R}^n \times \mathcal{B}_s(R)$ is open in $\mathbb{R}^n \times \mathbb{R}^n$ and $D_s(R) \cap (\mathbb{R}^n \times \mathbb{R}^n) = W_s$ (real points of $D_s(R)$: $\theta \in \mathbb{R}^n \subset T_s$ since $s > 0$).*
- *(b) $B(0,R)_s \cap \mathbb{R}^n = \mathcal{B}_s(R)$; $D_s(R)$ and $B(0,R)_s$ are open.*
- *(c) (Euclidean tubes) Let $P = \{J \in \mathbb{R}^n : |J| \le R/2\}$ and $0 < r_0 \le \min(s/2, R/8)$. If $I \in \mathbb{C}^n$ and $J \in P$ with $|I - J|_2 < r_0$, then $I \in B(0,R)_s$ and $|I - J|_\infty < s$. If moreover $I \in \mathbb{R}^n$ then $|I|_\infty < 5R/8 \le 3R/4$.*
- *(d) $T_\rho \subset T_s$ for $\rho \le s$; $T_s = \mathbb{C}^0$ is the universal singleton for $n = 0$ (no $n > 0$ hypothesis is needed anywhere in N-ZERO, N-QC).*

Source: Ours (Remark 1.9); BM11 p. 4 for the notation.

Proof. **Step S2.3: Real tube.** ($\subset$) $|I| \le |J| + |I - J| < R + s$. ($\supset$) If $|I| < R + s$ put $J = \frac{R}{R+s} I$; then $|J| < R$ and $|I - J| = \frac{s}{R+s}|I| < s$ (for $I = 0$ take $J = 0$).

Step S2.4: Openness and real points. $B(0,R)_s = \bigcup_{J \in B(0,R)} \{|I - J| < s\}$ is a union of open balls; T_s is open as a preimage of an open set; products of open sets are open; the rest is the definition of ι and $\mathcal{B}_s(R) = B(0,R)_s \cap \mathbb{R}^n$ by Step S2.3.

Step S2.5: Euclidean tubes. By (E2) $|I - J|_\infty \le |I - J|_2 < r_0 \le s/2 < s$, and $|J| < R$ since $J \in P$; so $I \in B(0,R)_s$. For real I: $|I|_\infty < R/2 + R/8$. □

2.3. **Holomorphy implies real smoothness.** Throughout this subsection $R, s > 0$, h satisfies `IsAnalyticIntegrable` $R\, s\, h$, f satisfies `IsAnalyticPerturbation` $R\, s\, \varepsilon\, f$ ($\varepsilon \ge 0$), $G(\theta, I) = h(I) + f(\theta, I)$ on $D_s(R)$, $H = H_{h,f}$ and $\iota : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{C}^n \times \mathbb{C}^n$ the componentwise inclusion. Complex partials are $\partial_{\theta_i} G(w) = DG(w)[(e_i, 0)]$, $\partial_{I_i} G(w) = DG(w)[(0, e_i)]$.

Lemma 2.3 (Real smoothness and real Hamiltonian field). *(a) G is holomorphic and C^∞ on $D_s(R)$; h is holomorphic and C^∞ on $B(0,R)_s$.*

- *(b) H is C^∞ on the open set $W_s = \mathbb{R}^n \times B(0, R+s)$ (Lemma 2.2); on W_s it is real and $H = G \circ \iota$, and for all orders $D^k H(z)[\zeta_1, \dots, \zeta_k] = D^k G(\iota z)[\iota\zeta_1, \dots, \iota\zeta_k] \in \mathbb{R}$. Likewise $h_\mathbb{R} = h \circ \iota$ is C^∞ on $\mathcal{B}_s(R)$, with $D^k h_\mathbb{R}(I)[a_1, \dots, a_k] = D^k h(I)[\iota a_1, \dots]$.*
- *(c) (Hamiltonian field) For $z = (\theta, I) \in W_s$: H is differentiable at z and $X_H(z) = \big((\partial_{I_i} G(\iota z))_i, (-\partial_{\theta_i} G(\iota z))_i\big) = X_{h+f}(\iota z)$, the complex Hamiltonian field of G at the real point ιz. The frequency map is $\omega_h(I)_i = \partial_i h(I) \in \mathbb{R}$ and $D^2 h_\mathbb{R}(I)[e_i, e_j] = \partial_i \partial_j h(I) = \partial_j \partial_i h(I) \in \mathbb{R}$ for $I \in \mathcal{B}_s(R)$.*
- *(d) (Periodicity) $H(\theta + k, I) = H(\theta, I)$ for $(\theta, I) \in W_s$, $k \in \mathbb{Z}^n$; consequently DH, $D^k H$ and X_H are 1-periodic in θ on W_s.*

(e) If $\varepsilon = 0$ then $f = 0$ on $D_s(R)$ and $H(\theta, I) = h_{\mathbb{R}}(I)$ on W_s.
Source: BM11 p. 4 (regularity of h on B_s); [K:bp:p1:holomorphic-regularity], [K:bp:p1:hamiltonian-real]*; ours for the packaging. Part 3 uses this as* [N:bpN:p3:holo-input].

Proof. **Step S2.6: Holomorphy.** h is complex differentiable on the open set $B(0,R)_s$ and f on $D_s(R)$ (Definitions 1.11(i), 1.12(i); openness Lemma 2.2(b)); the map $(\theta, I) \mapsto h(I)$ is holomorphic on $D_s(R)$ (composition with the linear projection; $B(0,R)_s$ is the second factor). So G is holomorphic on $D_s(R)$, hence C^∞ with holomorphic derivatives of all orders ([K:bp:p1:holomorphic-regularity]), which is (a); by (E3) G is C^∞ in the real sense.

Step S2.7: Reality. $\iota(W_s) \subset D_s(R)$ (Lemma 2.2(a)) and $W_s = \iota^{-1}(D_s(R))$ is open. For $z = (\theta, I) \in W_s$, $\operatorname{Im} h(I) = 0$ (Definition 1.11(iv), $I \in \mathcal{B}_s(R)$) and $\operatorname{Im} f(\theta, I) = 0$ (Definition 1.12(iii)); so $G(\iota z) \in \mathbb{R}$ and $H(z) = \operatorname{Re} G(\iota z) = G(\iota z)$. Likewise $h_{\mathbb{R}} = h \circ \iota$ on $\mathcal{B}_s(R)$.

Step S2.8: Derivatives of all orders. On the open set W_s, $H = G \circ \iota$ with ι $\mathbb{R}$-linear and G real-C^∞ (Step S2.6): chain rule (E3) gives $H \in C^\infty(W_s)$ and the displayed formula for $D^k H$, with values real because H is real-valued (the derivative of a real-valued function along real directions is a limit of real difference quotients). The same for $h_{\mathbb{R}}$.

Step S2.9: The field. By Definition 1.3, $X_H(z) = ((DH(z)[(0, e_i)])_i, (-DH(z)[(e_i, 0)])_i)$; by Step S2.8 these are $\partial_{I_i} G(\iota z)$ and $-\partial_{\theta_i} G(\iota z)$. This is [K:bp:p1:hamiltonian-real] (with $V = D_s(R)$, H agreeing with $\operatorname{Re} G$ on the real neighbourhood W_s). Taking $k = 1, 2$ for $h_{\mathbb{R}}$ gives $\omega_h(I)_i = Dh_{\mathbb{R}}(I)[e_i] = \partial_i h(I)$ and $D^2 h_{\mathbb{R}}(I)[e_i, e_j] = \partial_i \partial_j h(I)$, symmetric by [K:bp:p1:holomorphic-regularity] ([K:bp:p1:ext-schwarz]).

Step S2.10: Periodicity. For $(\theta, I) \in W_s$, $k \in \mathbb{Z}^n$ and real θ: $(\theta, I), (\theta + k, I) \in D_s(R)$ and $f(\theta + k, I) = f(\theta, I)$ (Definition 1.12(ii)), $h(I)$ does not depend on θ. So $H \circ \tau_k = H$ on W_s, $\tau_k(\theta, I) = (\theta + k, I)$, which maps W_s onto itself with $D\tau_k = \mathrm{id}$; differentiating (chain rule) gives $D^j H(\tau_k z) = D^j H(z)$ and hence $X_H(\tau_k z) = X_H(z)$.

Step S2.11: Zero perturbation. $|f(w)| \le 0$ on $D_s(R)$ gives $f = 0$ there, so $H = \operatorname{Re} h(I) = h_{\mathbb{R}}(I)$ on W_s. □

Lemma 2.4 (Ordered partials and first-order bounds)**.** *Let $U \subset \mathbb{R}^n$ be open with* `HasDerivativeBounds` *$U\, h\, M$ (Definition 1.13). Then for $I \in U$, $a, b \in \mathbb{R}^n$: $|\omega_h(I)_i| \le M$, $|\omega_h(I)|_\infty \le M$, $|\omega_h(I)|_2 \le \sqrt{n}\, M$, and $|D^2 h_{\mathbb{R}}(I)[e_i, e_j]| \le M$. The ordered partial $D^k h_{\mathbb{R}}(I)[e_{i_1}, \dots, e_{i_k}]$ ($k \le 3$) is invariant under permutations of $(i_1, \dots, i_k)$, so it equals the multi-index partial of the source.*
Source: BM11 p. 4 ($B(M)$); [K:bp:p1:ext-schwarz].

Proof. **Step S2.12: First order.** $k = 1$ in the definition gives $|Dh_{\mathbb{R}}(I)[e_i]| = |\omega_h(I)_i| \le M$ (Definition 1.2); then $|\omega|_2 \le \sqrt{n}|\omega|_\infty$ (Lemma 2.1).

Step S2.13: Symmetry. $h_{\mathbb{R}} \in C^3$ on the open set U: $D^2 h_{\mathbb{R}}(I)$ and $D^3 h_{\mathbb{R}}(I)$ are symmetric ([K:bp:p1:ext-schwarz] applied to $h_{\mathbb{R}}$, $Dh_{\mathbb{R}}$, $D^2 h_{\mathbb{R}}$), which is the permutation invariance; the entry bound is $k = 2$. □

Lemma 2.5 (Complex Hessian bound in Euclidean form)**.** *Let* `HasComplexHessianBound` *$R\, s\, h\, M_{\mathbb{C}}$. For $I \in B(0,R)_s$ let $Q(I) = (\partial_i \partial_j h(I))_{ij}$ (complex symmetric). Then $|Q(I)_{ij}| \le M_{\mathbb{C}}$ and $|x^\top Q(I) y| \le n M_{\mathbb{C}} |x|_2 |y|_2$ for $x, y \in \mathbb{C}^n$; hence the operator norm of $Q(I) : (\mathbb{C}^n, |\cdot|_2) \to (\mathbb{C}^n, |\cdot|_2)$ is $\le n M_{\mathbb{C}}$. For real $I \in \mathcal{B}_s(R)$ this matrix is the real Hessian $D^2 h_{\mathbb{R}}(I)[e_i, e_j]$ (Lemma 2.3(c)), and for real x, y: $|D^2 h_{\mathbb{R}}(I)[x, y]| \le n M_{\mathbb{C}} |x|_2 |y|_2$.*
Source: Pos93 p. 190 (Hessian bound, anchor POS190-HESS); the conversion factor n is ours (`docs/tracks/N/sources.md`).

Source gap: Pos93 p. 190 says that for the symmetric Q the Euclidean operator norm "amounts to the spectral radius norm". For complex symmetric matrices this is false ($Q = \left(\begin{smallmatrix} 1 & i \\ i & -1 \end{smallmatrix}\right)$ has $Q^2 = 0$). The frozen definition uses the operator norm of D^2h for the max norm, and so does this blueprint; no spectral-radius claim is used.

Proof. **Step S2.14: Entries.** $D^2h(I) = D(Dh)(I) \in \mathcal{L}(\mathbb{C}^n, \mathcal{L}(\mathbb{C}^n, \mathbb{C}))$ has norm $\le M_{\mathbb{C}}$ by hypothesis and $|e_i| = 1$ for the max norm, so $|\partial_i \partial_j h(I)| = |D^2h(I)[e_i, e_j]| \le M_{\mathbb{C}}$ (the order of arguments is immaterial by symmetry, [K:`bp:p1:holomorphic-regularity`]).

Step S2.15: Euclidean bound. Lemma 2.1(c). The operator-norm statement: $|Qx|_2 = \sup_{|y|_2=1} |\sum_i \overline{y}_i (Qx)_i| \le n M_{\mathbb{C}} |x|_2$ by the same estimate applied to $\overline{y}$. □

2.4. Quasi-convexity: the uniform Euclidean form.

Lemma 2.6 (Transverse coercivity gives conic coercivity (linear algebra core))**.** *Let $n \ge 1$, $B, q > 0$, $Q : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ bilinear (not necessarily symmetric) with $|Q[x, y]| \le B|x|_2|y|_2$, $\omega \in \mathbb{R}^n \setminus \{0\}$, and*

$$\omega \cdot v = 0 \implies Q[v, v] \ge q|v|_2^2 \qquad (v \in \mathbb{R}^n).$$

If $\ell' \ge 0$ and $\ell' \le |\omega|_2 \, q/(8(B + q))$, then for every $\xi \in \mathbb{R}^n$:

$$|\omega \cdot \xi| \le \ell' |\xi|_2 \implies Q[\xi, \xi] \ge \tfrac{5}{8} q |\xi|_2^2 \ \ (\ge \tfrac{q}{2} |\xi|_2^2).$$

Source: Ours; the computation is in `docs/tracks/N/sources.md` ("Uniform conversion"); all inequalities re-checked here.

Proof. **Step S2.16: Reduction to unit vectors.** Both hypothesis and conclusion are homogeneous (degrees 1 and 2) in ξ, and the case $\xi = 0$ is trivial; so let $|\xi|_2 = 1$.

Step S2.17: Decomposition. Put $e = \omega/|\omega|_2$, $t = \xi \cdot e$, $u = \xi - te$. Then $u \cdot \omega = 0$, $|u|_2^2 = 1 - t^2$, $|t| \le 1$, and $|t| = |\omega \cdot \xi|/|\omega|_2 \le \ell'/|\omega|_2 \le \kappa := q/(8(B + q)) < \frac{1}{8}$.

Step S2.18: Expansion. $Q[\xi, \xi] = Q[u, u] + tQ[u, e] + tQ[e, u] + t^2 Q[e, e]$. By hypothesis $Q[u, u] \ge q(1 - t^2)$; by the bound on Q and $|u|_2, |e|_2 \le 1$: $|Q[u, e]|, |Q[e, u]|, |Q[e, e]| \le B$. So $Q[\xi, \xi] \ge q - (q + B)t^2 - 2B|t|$.

Step S2.19: Numerics. $(q + B)t^2 \le (q + B)\kappa \cdot \kappa = \frac{q}{8}\kappa \le \frac{q}{8}$ and $2B|t| \le 2B\kappa = \frac{2Bq}{8(B+q)} \le \frac{q}{4}$ (as $B \le B + q$). Hence $Q[\xi, \xi] \ge q - \frac{q}{8} - \frac{q}{4} = \frac{5}{8} q$. □

Lemma 2.7 (Quasi-convexity in Pöschel's quantified form)**.** *Let $n \ge 1$, $R, s, m, M_{\mathbb{C}}, \omega_{\min} > 0$ and put*

$$q = \frac{m}{n}, \qquad B_c = n M_{\mathbb{C}} + q, \qquad \mu = \frac{q}{2}, \qquad \ell = \frac{\omega_{\min} \, q}{8(B_c + q)}.$$

Let h satisfy `IsAnalyticIntegrable` $R\, s\, h$ and `HasComplexHessianBound` $R\, s\, h\, M_{\mathbb{C}}$, `IsStrictlyQuasiConvex` $\mathcal{B}_s(R)\, h\, m$, and $\omega_{\min} \le |\omega_h(I)|_\infty$ for all $I \in B(0, R)$. Then for every $I \in B(0, R)$, writing $\omega = \omega_h(I)$, $Q = D^2 h_{\mathbb{R}}(I)$:

- *(a) $|Q[x, y]| \le B_c |x|_2 |y|_2$; $\omega \cdot v = 0 \Rightarrow Q[v, v] \ge q|v|_2^2$;*
- *(b) (Pöschel's alternative,* conditional*) for every $\xi \in \mathbb{R}^n$: $|\omega \cdot \xi| \le \ell |\xi|_2 \implies Q[\xi, \xi] \ge \mu |\xi|_2^2$; equivalently, either $|\omega \cdot \xi| > \ell |\xi|_2$ or $Q[\xi, \xi] \ge \mu |\xi|_2^2$ (Pos93 p. 190, "ℓ, m-quasi-convex" with $m := \mu$);*
- *(c) $|\omega|_2 \ge |\omega|_\infty \ge \omega_{\min}$ and $|\omega|_2 \le \sqrt{n} |\omega|_\infty$.*

The constants q, B_c, μ, ℓ depend only on $(n, m, M_{\mathbb{C}}, \omega_{\min})$. In Pöschel's normalisation ($\tilde{h} = 2\pi h$, Lemma 2.8) the same holds with $(\tilde{\ell}, \tilde{\mu}, \tilde{B}_c, \tilde{\omega}, \tilde{Q}) = 2\pi(\ell, \mu, B_c, \omega, Q)$. Warning. *Full positive definiteness $Q[\xi, \xi] \ge \mu |\xi|_2^2$ for all ξ is false in general and is not claimed: for $h(I) = I_1 + \frac{1}{2}(I_2^2 + I_3^2)$ one has $\omega = e_1 \ne 0$, QC holds on every bounded set, but $Q[e_1, e_1] = 0$. The estimate (b) is only needed (and only true) in the cone $|\omega \cdot \xi| \le \ell |\xi|_2$.*

Source: Pos93 p. 190 (anchors POS190-QC, POS190-HESS, POS191-LOW; I read the scan, p. 190, ℓ, m-quasi-convexity, hypothesis $r_0 \le 4\ell/m$); conversion: `docs/tracks/N/sources.md`*, "Uniform conversion"; BM11 p. 4 for $QC(m)$. Review N05.*

Proof. **Step S2.20: Regularity and Hessian.** Since $B(0,R) \subset \mathcal{B}_s(R)$, $I \in \mathcal{B}_s(R)$; Q is the real Hessian of Lemma 2.5, so $|Q[x,y]| \le nM_{\mathbb{C}}|x|_2|y|_2 \le B_c|x|_2|y|_2$, the first half of (a).

Step S2.21: Transverse coercivity in the Euclidean norm. Definition 1.14 at $I \in \mathcal{B}_s(R)$ gives $\omega \cdot v = 0 \Rightarrow Q[v,v] \ge m|v|_\infty^2 \ge \frac{m}{n}|v|_2^2 = q|v|_2^2$ (Lemma 2.1(a)). Also $\omega \neq 0$.

Step S2.22: Frequency size. (c): $|\omega|_\infty \ge \omega_{\min}$ on $B(0,R)$ by hypothesis, $|\omega|_2 \ge |\omega|_\infty$; $|\omega|_2 \le \sqrt{n}|\omega|_\infty$ by Lemma 2.1.

Step S2.23: Conic coercivity. Apply Lemma 2.6 with $B = B_c$, q and $\ell' = \ell$: $\ell = \omega_{\min} q/(8(B_c + q)) \le |\omega|_2 q/(8(B_c+q))$ by (c). The conclusion is $Q[\xi,\xi] \ge \frac{5}{8}q|\xi|_2^2 \ge \mu|\xi|_2^2$.

Step S2.24: Scaling. Multiplying h by 2π multiplies ω, Q, B_c and the three of μ, q, ℓ by 2π, and the implications are homogeneous, so they persist. □

2.5. **Normalisations.**

Lemma 2.8 (Period conversion)**.** *Let $\Lambda(\theta, I) = (2\pi\theta, I)$ and, for $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ and h, f complex, define*

$$\tilde{H}(\varphi, I) = 2\pi H(\varphi/2\pi, I), \qquad \tilde{h} = 2\pi h, \qquad \tilde{f}(\varphi, I) = 2\pi f(\varphi/2\pi, I),$$

so $\tilde{H} = H_{\tilde{h},\tilde{f}}$ when $H = H_{h,f}$, and $\tilde{f}$ is $2\pi\mathbb{Z}^n$-periodic in φ. Then:

- *(a) (Dynamics) For $U \subset \mathbb{R}^n$, $J \subset \mathbb{R}$: z satisfies* `IsSolutionOn` $H\,U\,J\,z$ *iff $\Lambda \circ z$ satisfies* `IsSolutionOn` $\tilde{H}\,U\,J$ *(same time, same actions). In particular $|(\Lambda z)_2 - I_0| = |z_2 - I_0|$.*
- *(b) (Quantities) $\omega_{\tilde{h}} = 2\pi\omega_h$, $D^2\tilde{h}_{\mathbb{R}} = 2\pi D^2 h_{\mathbb{R}}$, and so:* `IsStrictlyQuasiConvex` $U\,h\,m \Rightarrow$ `IsStrictlyQuasiConvex` $U\,\tilde{h}\,2\pi m$*; $B(M) \Rightarrow$ coordinate partials of $\tilde{h}_{\mathbb{R}}$ of order $1,2,3$ bounded by $2\pi M$; the complex Hessian bound $M_{\mathbb{C}} \to 2\pi M_{\mathbb{C}}$; $\omega_{\min} \to 2\pi\omega_{\min}$; Euclidean sizes of ω: $|\tilde{\omega}|_2 = 2\pi|\omega|_2$. Time is unchanged.*
- *(c) (Domains) $\tilde{f}$ is holomorphic on $\{|\mathrm{Im}\,\varphi| < 2\pi s\} \times B(0,R)_s$, with $|\tilde{f}| \le 2\pi\varepsilon$ if $|f| \le \varepsilon$ on $D_s(R)$; the strip T_ρ (ρ in θ) corresponds to the strip of width $2\pi\rho$ in φ.*
- *(d) (Fourier) $\tilde{f}(\varphi, I) = \sum_k \hat{f}_k(I) e^{ik\cdot\varphi}$ with the same coefficients $\hat{f}_k(I) = \int_{[0,1]^n} f(\theta, I) e^{-2\pi i k\cdot\theta}\, d\theta$ times 2π; the coefficient radius s' of Pöschel's weight $e^{|k|_1 s'}$ equals $2\pi\rho$.*

Source: `docs/tracks/N/sources.md` *("For the period conversion"); Pos93 p. 189 (set-up: $\theta \in \mathbb{R}^n/2\pi\mathbb{Z}^n$, strip W_{s_0} of width s_0 in the max norm of $\mathrm{Im}\,\theta$).*

Proof. **Step S2.25: Chain rule.** $\tilde{H} = 2\pi\, H \circ \Lambda^{-1}$; Λ is a linear isomorphism, so $\tilde{H}$ is differentiable at Λz iff H is at z, with $\partial_{I_i}\tilde{H}(\Lambda z) = 2\pi\partial_{I_i}H(z)$ and $\partial_{\varphi_i}\tilde{H}(\Lambda z) = \partial_{\theta_i}H(z)$.

Step S2.26: Vector field. Hence $X_{\tilde{H}}(\Lambda z) = (2\pi\partial_I H(z), -\partial_\theta H(z)) = (2\pi\dot{\theta}, \dot{I}) = \frac{d}{dt}\Lambda z$ when $\dot{z} = X_H(z)$, and conversely; the condition $z(t)_2 \in U$ is unchanged. This is (a). (Rescaling the angle without the factor 2π in H would change time.)

Step S2.27: Scalars. (b) is linearity of $h \mapsto \omega_h, D^2 h_{\mathbb{R}}, D^k h_{\mathbb{R}}$ and of the operator norm. (c),(d) are the substitution $\varphi = 2\pi\theta$, $|\mathrm{Im}\,\varphi| = 2\pi|\mathrm{Im}\,\theta|$, and $e^{ik\cdot\varphi} = e^{2\pi i k\cdot\theta}$. □

Lemma 2.9 (From sup norm on a strip to the weighted Fourier norm)**.** *Let $u : T_\rho \to \mathbb{C}$ be holomorphic, bounded and 1-periodic, $\|u\|_\rho = \sup_{T_\rho}|u|$, and $0 < \delta < \rho$. Then*

$$\sum_{k\in\mathbb{Z}^n} |\hat{u}_k|\, e^{2\pi|k|_1(\rho-\delta)} \;\le\; \coth(\pi\delta)^n\, \|u\|_\rho,$$

the left side is Pöschel's weighted norm with weight $e^{|k|_1 s'}$, $s' = 2\pi(\rho-\delta)$, and the factor is $\coth^n \sigma$ with $\sigma = \pi\delta$ (Pos93 p. 190: $|u|_{D,r,s} \le \coth^n \sigma\, |u|_{D,r,s+2\sigma}$, where the right side is the sup norm on the strip of width $s+2\sigma$ in φ, i.e. $\rho = (s'+2\sigma)/2\pi$). Conversely $\|u\|_\rho \le \sum_k |\hat u_k| e^{2\pi|k|_1\rho}$. For I-dependent $u(\theta, I)$ the statement holds for each fixed I with the same constant, so the sup over I in the weighted norm is $\le \coth(\pi\delta)^n \sup_{\theta,I}|u|$.
Source: Pos93 p. 190 (the displayed relation between the weighted and sup norms); frozen-convention statement: [K:bp:p1:summation], [K:bp:p1:decay], [K:bp:p1:exp-sums]*(b).*

Proof. **Step S2.28: Decay.** For $\rho' \in (\rho-\delta, \rho)$, [K:bp:p1:decay] gives $|\hat u_k| \le e^{-2\pi|k|_1\rho'}\|u\|_\rho$; hence $|\hat u_k| e^{2\pi|k|_1(\rho-\delta)} \le e^{-2\pi|k|_1(\rho'-\rho+\delta)}\|u\|_\rho$, and letting $\rho' \uparrow \rho$ (termwise, then summing, or directly [K:bp:p1:summation], converse part) the bound is $e^{-2\pi\delta|k|_1}\|u\|_\rho$ per k.

Step S2.29: Sum. $\sum_{k\in\mathbb{Z}^n} e^{-2\pi\delta|k|_1} = \coth(\pi\delta)^n$ ([K:bp:p1:exp-sums](b) with $x = 2\pi\delta$). The case with parameters is pointwise in I. □

2.6. **The dictionary of constants.**

Lemma 2.10 (Constants dictionary for N-HEAD-C)**.** *Let $n \ge 3$ (only $n \ge 2$ is used by Pöschel) and $R, s, M, m, M_{\mathbb{C}}, \omega_{\min} > 0$, $a = \frac{1}{2n}$. Define successively*

$$q = \tfrac{m}{n},\ B_c = nM_{\mathbb{C}} + q,\ \mu = \tfrac{q}{2},\ \ell = \tfrac{\omega_{\min} q}{8(B_c+q)}, \qquad r_0 = \min\Big(\tfrac{s}{2}, \tfrac{R}{8}, \tfrac{2\ell}{\mu}, \tfrac{4\omega_{\min}}{\mu}\Big), \qquad A = \tfrac{11B_c}{\mu},$$

$$s_0 = \pi s, \quad \sigma = \tfrac{\pi s}{4}, \quad F = 2\pi \coth^n\big(\tfrac{\pi s}{4}\big), \quad \varepsilon_* = \frac{2\pi\mu\, r_0^2}{2^{10} A^{2n}},$$

$$C = \tfrac{r_0}{A}\Big(\tfrac{F}{\varepsilon_*}\Big)^a, \quad c = \tfrac{s_0}{6}\Big(\tfrac{\varepsilon_*}{F}\Big)^a, \quad T = \frac{A^2 s_0}{4\pi n M},$$

$$\varepsilon_0 = \min\Big(\tfrac{\varepsilon_*}{F},\ \big(\tfrac{R}{4C}\big)^{2n}\Big).$$

Pöschel's data (his normalisation, for $\tilde h = 2\pi h$): $m_P = 2\pi\mu$, $\ell_P = 2\pi\ell$, $M_P = 2\pi B_c$, $A_P = 11M_P/m_P = A$, $\varepsilon_{0,P} = m_P r_0^2/(2^{10}A_P^{2n}) = \varepsilon_*$, $R_0 = r_0/A$, and the set $P = \{J \in \mathbb{R}^n : |J| \le R/2\}$. Then:

- *(a) All constants are positive and finite and depend only on $(n, R, s, M, m, M_{\mathbb{C}}, \omega_{\min})$ — not on $h, f, \varepsilon, \theta_0, I_0$, nor on the bound A of Definition 1.11(iii). Order of choice: $q, B_c, \mu, \ell, r_0, A$; then s_0, σ, F; ε_*; C, c; T; ε_0. And $C\varepsilon_0^a \le R/4$, $0 < \varepsilon_0 \le \varepsilon_*/F$.*
- *(b) $r_0 \le 4\ell_P/m_P$ $(= 4\ell/\mu)$; $A \ge 22$; $R_0 = r_0/A \le r_0/22 \le R/176$.*
- *(c) (Tubes) With $V_{r_0}P = \{I \in \mathbb{C}^n : \exists J \in P,\ |I-J|_2 < r_0\}$ and $U_{r_0}P = V_{r_0}P \cap \mathbb{R}^n$: $V_{r_0}P \subset B(0,R)_s$ and $U_{r_0}P \subset B(0, 5R/8) \subset B(0,R)$.*
- *(d) (Hypotheses of Theorem 1) If h satisfies (H1)–(H5) of Theorem 1.19, then for $I \in U_{r_0}P$: $\tilde h$ is ℓ_P, μ_P-quasi-convex (Lemma 2.7) with $\mu_P = 2\pi\mu$; the complex Hessian operator norm (Euclidean) of $\tilde h$ on $V_{r_0}P$ is $\le M_P$ (Lemma 2.5); Pöschel's m, M are m_P, M_P and his hypothesis $r_0 \le 4\ell_P/m_P$ holds.*
- *(e) (Perturbation) If f satisfies (H6) with ε, then Pöschel's norm of $\tilde f$ on $V_{r_0}P \times$(strip of width s_0) is $\le F\varepsilon$; and $F\varepsilon \le \varepsilon_{0,P}$ if $\varepsilon \le \varepsilon_0$.*
- *(f) (Exceptional branch is empty) For $I_0 \in B(0, R/2)$: $|\tilde\omega(I_0)|_2 \ge 2\pi\omega_{\min} > \pi\omega_{\min} \ge m_P r_0/8$.*
- *(g) (Time prefactor) For $I_0 \in B(0, R/2)$, $\Omega_0 = \sup_{|I-I_0|_2 \le R_0} |\tilde\omega(I)|_2 \le 2\pi\sqrt{n}M \le 2\pi n M$, so Pöschel's $T_0 = A_P^2 s_0/\Omega_0 \ge 2T$.*
- *(h) (Conclusion shape) Pöschel's $R_0(\varepsilon_P/\varepsilon_{0,P})^a$ and $T_0 \exp\big(\frac{s_0}{6}(\varepsilon_{0,P}/\varepsilon_P)^a\big)$ with $\varepsilon_P = F\varepsilon$ are $C\varepsilon^a$ and $T_0 \exp(c\varepsilon^{-a})$, and $T\exp(c\varepsilon^{-a}) \le \frac12 T_0 \exp(c\varepsilon^{-a})$.*

Source: `docs/tracks/N/sources.md` "Uniform conversion" (all constants), checked against Pos93 p. 190 (hypotheses, $A = 11M/m$, $\varepsilon_0 = mr_0^2/(2^{10}A^{2n})$, $r_0 \le 4\ell/m$), p. 191 (parameters a, R_0, T_0, exceptional branch) and p. 204 (assembly: $K = (\varepsilon_0/\varepsilon)^a$). Pos93 requires $\|f_\varepsilon\| \le \varepsilon$ (p. 190) with $\varepsilon \le \varepsilon_{0,P}$.

Source gap: Pos93 states no theorem about the $\omega_{\min}$-conversion and the frozen-to-source dictionary: it exists only in `sources.md` *and here (Part 0* `bpN:p0:head`*). The choice of T with the factor $1/2$ is slack, not needed by the source.*

Proof. **Step S2.30: Positivity and dependence.** Each constant is a positive finite expression in earlier ones (denominators: $n, \mu, 8(B_c+q), A, F, \varepsilon_*, 4C, 4\pi nM$, all >0); $\varepsilon_0 > 0$ as a minimum of positive numbers, and $C\varepsilon_0^a \le C \cdot \frac{R}{4C} = R/4$ since $((R/4C)^{2n})^{1/(2n)} = R/(4C)$. The list of inputs in each formula contains only $n, R, s, M, m, M_{\mathbb{C}}, \omega_{\min}$. This is (a).

Step S2.31: Size relations. $B_c = nM_{\mathbb{C}} + q \ge q = 2\mu$ so $A = 11B_c/\mu \ge 22$. $4\ell_P/m_P = 4\ell/\mu \ge 2\ell/\mu \ge r_0$, and $R_0 = r_0/A \le r_0/22 \le R/176$. This is (b).

Step S2.32: Tubes. $r_0 \le \min(s/2, R/8)$: Lemma 2.2(c) gives (c).

Step S2.33: Quasi-convexity and Hessian. For (d): $U_{r_0}P \subset B(0,R)$ by (c), so Lemma 2.7(b) applies at every point of $U_{r_0}P$ and, after Lemma 2.8(b), gives the (ℓ_P, μ_P)-alternative for $\tilde{h}$. For $I \in V_{r_0}P \subset B(0,R)_s$ the Euclidean operator norm of $\tilde{Q}(I)$ is $\le 2\pi nM_{\mathbb{C}} \le 2\pi B_c = M_P$ (Lemma 2.5). Pöschel's $A = 11M/m$ with $M = M_P$ and $m = \mu_P = m_P$ is $11B_c/\mu = A$.

Step S2.34: Perturbation. By (H6) $|f| \le \varepsilon$ on $D_s(R) \supset T_{3s/4} \times V_{r_0}P$ ($V_{r_0}P \subset B(0,R)_s$, (c)). For each fixed $I \in V_{r_0}P$ apply Lemma 2.9 to $u = f(\cdot, I)$ with $\rho = 3s/4$, $\delta = s/4$: $\sum_k |\hat{f}_k(I)| e^{2\pi|k|_1 s/2} \le \coth(\pi s/4)^n \varepsilon$. Multiplying by 2π (Lemma 2.8(d)) and noting $2\pi \cdot s/2 = \pi s = s_0$ gives $\le F\varepsilon$. (In Pöschel's terms $s_0 + 2\sigma = 3\pi s/2 = 2\pi \cdot \frac{3s}{4}$.) $\varepsilon \le \varepsilon_*/F$ gives $F\varepsilon \le \varepsilon_* = \varepsilon_{0,P}$. This is (e).

Step S2.35: Exceptional branch. $I_0 \in B(0, R/2) \subset B(0,R)$ so $|\omega(I_0)|_2 \ge \omega_{\min}$ (Lemma 2.7(c)) and $|\tilde{\omega}|_2 = 2\pi|\omega|_2$. And $m_P r_0/8 = 2\pi\mu r_0/8 \le 2\pi\mu \cdot \frac{4\omega_{\min}}{8\mu} = \pi\omega_{\min}$ by $r_0 \le 4\omega_{\min}/\mu$. This is (f).

Step S2.36: Time. The Euclidean ball $\{|I - I_0|_2 \le R_0\}$ with $R_0 \le r_0$ lies in $\{|I|_\infty < R/2 + R_0\} \subset B(0, 5R/8)$ ($|I - I_0|_\infty \le |I - I_0|_2$), inside $\mathcal{B}_s(R)$ where Lemma 2.4 gives $|\omega(I)|_2 \le \sqrt{n}M$. So $\Omega_0 \le 2\pi\sqrt{n}M \le 2\pi nM$ and $T_0 = A^2 s_0/\Omega_0 \ge A^2 s_0/(2\pi nM) = 2T$. (g).

Step S2.37: Exponent and constants. $R_0(F\varepsilon/\varepsilon_*)^a = \frac{r_0}{A}(F/\varepsilon_*)^a\varepsilon^a = C\varepsilon^a$ and $\frac{s_0}{6}(\varepsilon_*/(F\varepsilon))^a = c\varepsilon^{-a}$. (h) follows from $T \le T_0/2$. □

Remark 2.11 (Forward use and an unused constant). Part 10 chooses C, T, c, ε_0 as in Lemma 2.10, verifies Pos93's hypotheses by (d),(e) and excludes the exceptional branch by (f). The time prefactor T enters only through (g); the remaining slack $\varepsilon_0 \le (R/4C)^{2n}$ gives $C\varepsilon^a \le R/4$ so that the a-priori confinement keeps $I(t) \in B(0, R/2 + R/4) = B(0, 3R/4)$ (continuation, [N:`bpN:p3:continuation`]). The value $M_{\mathbb{C}}$ enters only through B_c; M only through T.

3. N-QC and N-ZERO

This part proves two frozen theorems completely: N-QC (Theorem 1.21, the nonsingular isoenergetic derivative) and N-ZERO (Theorem 1.20, the exact integrable dynamics). Neither uses the normal-form machinery. The linear-algebra core of N-QC is stated in a reusable form (Lemma 3.1, cited by track I as the isoenergetic gate, [N:`bpN:p2:qc-kernel`]). Frozen conventions throughout (max norms, period 1); Euclidean structure appears only as the dot product $\omega \cdot v = \sum \omega_i v_i$.

3.1. N-QC.

Lemma 3.1 (Kernel core: bordered Hessian is nonsingular)**.** *Let $n \ge 1$, $\omega \in \mathbb{R}^n$, A a real $n \times n$ matrix (not necessarily symmetric, no norm bound), $\lambda \ne 0$, and suppose*

$$\omega \cdot v = 0,\ v \ne 0 \implies v^\top A v \ne 0 \qquad \text{and} \qquad \omega \ne 0.$$

Let $L : \mathbb{R}^n \times \mathbb{R} \to \mathbb{R} \times \mathbb{R}^n$, $L(v,u) = (\omega \cdot v,\ u\,\omega + \lambda A v)$. Then $\ker L = 0$, hence L is bijective. In particular this holds if $\lambda > 0$, $\omega \ne 0$ and $v^\top A v \ge m|v|^2$ for some norm and $m > 0$ whenever $\omega \cdot v = 0$. (For $n = 0$ no $\omega \ne 0$ exists; N-QC is then vacuous, Remark 3.2.)

Source: BM11 Lemma 3.3 and its proof, pp. 9–10 (kernel computation); linear algebra (E1).

Proof. **Step S3.1: Kernel equations.** Let $L(v,u)=0$: $\omega\cdot v=0$ and $u\omega+\lambda Av=0$.

Step S3.2: Pairing with v. Dot the second equation with v: $u(\omega\cdot v)+\lambda v^\top Av=0$, so $\lambda v^\top Av=0$, i.e. $v^\top Av=0$ as $\lambda\neq 0$. Since also $\omega\cdot v=0$, the hypothesis forces $v=0$.

Step S3.3: The scalar. With $v=0$: $u\omega=0$; as $\omega\neq 0$, $u=0$.

Step S3.4: Bijectivity. L is a linear map between spaces of dimension $n+1$ with trivial kernel, hence bijective by (E1). □

Remark 3.2 ($n=0$). For $n=0$ the space $\mathbb{R}^0$ is a point, so $\omega=0$ and Definition 1.14 requires $\omega_h(I)\neq 0$: no I satisfies it. Hence for $n=0$ the hypotheses of N-QC are never met with $I\in U$ and the theorem is vacuous; no restriction $n\geq 1$ is added to it (review N03).

Lemma 3.3 (The derivative of Ψ)**.** *Let $U\subset\mathbb{R}^n$ be open, h with $h_{\mathbb{R}}\in C^2(U)$, $I\in U$, $\lambda\in\mathbb{R}$, $\Psi(I',\lambda')=(h_{\mathbb{R}}(I'),\lambda'\omega_h(I'))$. Then Ψ is differentiable at (I,λ) (so `fderiv` is the genuine derivative; Remark 1.18(e)) and*

$$D\Psi(I,\lambda)[v,u]=\big(\omega\cdot v,\ u\,\omega+\lambda Qv\big),\qquad \omega=\omega_h(I),\ Q_{ij}=D^2h_{\mathbb{R}}(I)[e_i,e_j]=Q_{ji}.$$

Source: BM11 p. 10; (E4), [K:`bp:p1:ext-schwarz`].

Proof. **Step S3.5: First component.** $(I',\lambda')\mapsto h_{\mathbb{R}}(I')$ is differentiable at (I,λ) ($h_{\mathbb{R}}$ is C^2 on the open set $U\ni I$) with derivative $[v,u]\mapsto Dh_{\mathbb{R}}(I)[v]=\sum_i v_i Dh_{\mathbb{R}}(I)[e_i]=\omega\cdot v$.

Step S3.6: The frequency map is C^1. $\omega_h(I')_i=Dh_{\mathbb{R}}(I')[e_i]$ is the composition of $I'\mapsto Dh_{\mathbb{R}}(I')\in\mathcal{L}(\mathbb{R}^n,\mathbb{R})$ (which is C^1 on U since $h_{\mathbb{R}}\in C^2(U)$ and U is open) with the continuous linear evaluation $\ell\mapsto\ell(e_i)$. So ω_h is differentiable on U with $D\omega_h(I)[v]_i=D(Dh_{\mathbb{R}})(I)[v][e_i]=D^2h_{\mathbb{R}}(I)[v,e_i]$.

Step S3.7: Symmetry. $D^2h_{\mathbb{R}}(I)$ is symmetric ([K:`bp:p1:ext-schwarz`]), so $D\omega_h(I)[v]=Qv$ with Q the symmetric Hessian matrix.

Step S3.8: Product. By the product rule (E4), $(I',\lambda')\mapsto\lambda'\omega_h(I')$ is differentiable at (I,λ) with derivative $[v,u]\mapsto u\,\omega+\lambda Qv$. Both components are differentiable, so Ψ is, with the stated derivative. □

Theorem 3.4 (N-QC)**.** *Theorem 1.21 holds: for `IsStrictlyQuasiConvexU h m`, $I\in U$, $\lambda>0$, the map $D\Psi(I,\lambda)$ is bijective.*
Frozen Lean declaration: `KAM.Nekhoroshev.quasiConvex_isoenergetic_derivative`
Source: BM11 Lemma 3.3 and proof, pp. 9–10.

Proof. **Step S3.9: Unpack the hypothesis.** Definition 1.14: $m>0$, U open, $h_{\mathbb{R}}\in C^2(U)$, $\omega:=\omega_h(I)\neq 0$, and $\omega\cdot v=0\Rightarrow m|v|^2\leq D^2h_{\mathbb{R}}(I)[v,v]$ (max norm on v). Note $n\geq 1$ because $\omega\neq 0$ (Remark 3.2).

Step S3.10: The derivative. By Lemma 3.3, $D\Psi(I,\lambda)=L$ of Lemma 3.1 with $A=Q$ the Hessian matrix, $v^\top Qv=D^2h_{\mathbb{R}}(I)[v,v]$ (bilinearity).

Step S3.11: Positivity on $\omega^\perp$. For $\omega\cdot v=0$, $v\neq 0$: $v^\top Qv\geq m|v|_\infty^2>0$, since $m>0$ and $|v|_\infty>0$. So the hypothesis of Lemma 3.1 holds (with $\lambda>0$).

Step S3.12: Conclusion. Lemma 3.1 gives bijectivity of $L=D\Psi(I,\lambda)=\texttt{fderiv}\,\mathbb{R}\,\Psi\,(I,\lambda)$. □

Remark 3.5 (Reusable form for track I). Lemma 3.1 needs no norm, no symmetry of A and no uniform constant: any quasi-convexity-type hypothesis ("$v^\top Av > 0$ on $\omega^\perp \setminus 0$") together with $\omega \neq 0$ and $\lambda \neq 0$ gives nonsingularity of the bordered matrix $\begin{pmatrix} \omega^\top & 0 \\ \lambda A & \omega \end{pmatrix}$. A quantitative inverse bound is not asserted by N-QC and is not provided here; it would follow, if needed, from the compactness of $\{v : |v|_2 = 1,\ \omega \cdot v = 0\}$ for a fixed (I, λ) but is not uniform in (I, λ).

3.2. **N-ZERO.**

Lemma 3.6 (The field at the integrable orbit)**.** *Let $R, s > 0$, `IsAnalyticIntegrable R s h`, `IsAnalyticPerturbation R s 0 f`, $H = H_{h,f}$, and $W_s = \mathbb{R}^n \times B(0, R+s)$. Then $H(\theta, I) = h_{\mathbb{R}}(I)$ on W_s; H is differentiable at every $z = (\theta, I) \in W_s$ with $DH(z)[(a,b)] = Dh_{\mathbb{R}}(I)[b]$ and*

$$X_H(\theta, I) = (\omega_h(I), 0).$$

Source: BM11 p. 1 ($H = h(I)$ when $f = 0$); ours.

Proof. **Step S3.13: f vanishes.** Definition 1.12(iv) with $\varepsilon = 0$: $|f| \le 0$ on $D_s(R)$, so $f = 0$ there. For $z = (\theta, I) \in W_s$, $(\theta, I) \in D_s(R)$ (Lemma 2.2(a)), so $H(z) = \operatorname{Re} h(I) = h_{\mathbb{R}}(I)$ (Lemma 2.3(e)).

Step S3.14: Differentiability. W_s is open and on it $H = h_{\mathbb{R}} \circ \mathrm{pr}_2$ with $h_{\mathbb{R}} \in C^3(\mathcal{B}_s(R))$ (Definition 1.11(v)), $\mathcal{B}_s(R) = B(0, R+s)$ open. So H is differentiable at each point of W_s with $DH(z)[(a,b)] = Dh_{\mathbb{R}}(I)[b]$ (chain rule with the continuous linear projection). The point-wise hypotheses of the totality remark 1.18 are met.

Step S3.15: Field. By Definition 1.3, $\partial_{I_i} H = Dh_{\mathbb{R}}(I)[e_i] = \omega_h(I)_i$ (Definition 1.2) and $\partial_{\theta_i} H = DH(z)[(e_i, 0)] = 0$. □

Theorem 3.7 (N-ZERO)**.** *Theorem 1.20 holds.*
Frozen Lean declaration: `KAM.Nekhoroshev.zero_perturbation`
Source: Ours (supplement to BM11 Theorem 2.1, p. 5).

Proof. **Step S3.16: Initial value and action.** $\gamma(t) = (\theta_0 + t\omega_h(I_0), I_0)$ (Definition 1.17) gives $\gamma(0) = (\theta_0, I_0)$ and $\gamma(t)_2 = I_0$ for all t: the first and third conclusions.

Step S3.17: The action stays in $B(0,R)$. $I_0 \in B(0, R/2) \subset B(0, R)$ since $R > 0$ (max norm: $|I_0| < R/2 < R$).

Step S3.18: Differentiability of H along the orbit. $\gamma(t) \in \mathbb{R}^n \times B(0,R) \subset W_s$ (Lemma 2.2(a): $B(0,R) \subset B(0, R+s)$). By Lemma 3.6 H is differentiable at $\gamma(t)$ for each t.

Step S3.19: Derivative of the orbit. γ is affine in t, so differentiable at every $t \in \mathbb{R}$ (two-sided) with $\gamma'(t) = (\omega_h(I_0), 0)$.

Step S3.20: The equation. Lemma 3.6 gives $X_H(\gamma(t)) = (\omega_h(I_0), 0) = \gamma'(t)$ for every $t \in \mathbb{R}$. With Steps S3.17–S3.19, every clause of Definition 1.16 holds with $U = B(0,R)$ and $J = \mathbb{R}$. □

Remark 3.8 (Scope). N-ZERO uses neither the holomorphy of Part 1 (beyond the stated clauses of the definitions) nor the ODE layer of Part 3; Part 3 supplies uniqueness if ever required for $\varepsilon = 0$ (γ is the unique solution by [N:`bpN:p3:gen-uniqueness`]), which is not asserted by the frozen statement. The case $\varepsilon = 0$ is excluded from N-HEAD-C by (H7).

4. Hamiltonian ODE infrastructure (shared by tracks I, N, P)

This part is the single ODE layer of the three blueprints I, N, P. It has two levels. The *general layer* (§4.2–§4.6) is about an autonomous vector field F of class C^1 on an open subset W of a finite-dimensional real normed space E: local existence, uniqueness on *every* interval, maximal solutions, continuation (compact sets, and closed sets with uniform bounds), first exit from an open set.

The *Hamiltonian layer* (§4.7) adds $F = X_H$: smoothness of X_H, energy conservation, translation in the angles. §4.8 shows that the frozen solution predicates of the three tracks (`IsSolutionOn`, `IsHamiltonianSolutionOn`, `IsRestrictedSolutionOn`, `IsNewtonSolution`, `IsFixedSolution`) are instances of the general predicate, and §4.9 gives the six track-N results of `PLAN.md`. Tracks I and P cite the general results as [N:`bpN:p3:gen-...`].

4.1. **Conventions and external facts.**

- E is a finite-dimensional real normed space (hence complete and proper: closed bounded sets are compact), $F : E \to E$ a *total* function, $W \subset E$ open. In the applications $E = \mathbb{R}^n \times \mathbb{R}^n$ with the max norm $|(\theta, I)| = \max(|\theta|, |I|)$ (the product norm of the Lean text, equal to the max norm on the $2n$ coordinates), or $E = (\mathbb{R}^2)^N \times (\mathbb{R}^2)^N$ for the Newtonian problems.
- F is C^1 *on* W means `ContDiffOn` $\mathbb{R}$ 1 F W; as W is open this gives $F \in C^1$ at each point of W, and no information off W. Off W the values of F never enter (Lean totality, below).
- An *interval* is an order-connected subset of $\mathbb{R}$ (Lean `OrdConnected`); open, closed, half-open, bounded or not, possibly a single point. $\overline{B}(x, \delta)$ is the closed ball.
- All derivatives $\dot{z}(t)$ in solution predicates are two-sided (Lean `HasDerivAt`) at every $t \in J$, including endpoints of J, exactly as in the frozen predicates ("z is defined on $\mathbb{R}$").

External facts. Mathlib-level statements, with the declarations that exist (checked in `Mathlib/Analysis/ODE/`); what is *not* in Mathlib is listed after them.

(E1) *Picard–Lindelöf, quantitative form.* Let $F : E \to E$, $x \in E$, $a, M, L \geq 0$ with $\|F(y)\| \leq M$ and F L-Lipschitz on $\overline{B}(x, a)$, $T \geq 0$ with $MT \leq a$. Then there is $\alpha : \mathbb{R} \to E$ with $\alpha(t_0) = x$ and α has derivative $F(\alpha(t))$ within $[t_0 - T, t_0 + T]$ at each t of that interval. (Mathlib: the root-namespace structure `IsPicardLindelof` $f\, t_0\, x_0\, a\, r\, L\, K$ of `Analysis/ODE/PicardLindelof.lean`, with $r = 0$ and a time-independent field, and `IsPicardLindelof.exists_eq_forall_mem_Icc_hasDerivWithinAt`$_0$ (Unicode subscript zero) of `Analysis/ODE/ExistUnique.lean`; Mathlib's norm bound is the parameter L (our M) and its Lipschitz parameter is K (our L); the time constraint is the field `mul_max_le`: $M \max(t_{\max} - t_0, t_0 - t_{\min}) \leq a - r$. Parameters are `NNReal`. The contraction on iterates means no smallness of LT is needed.) Exists.

(E2) *Uniqueness with a Lipschitz constant on a set.* If $S \subset E$ and F is L-Lipschitz on S, $f, g : [a, b] \to E$ are continuous, have right derivative $F(f(t))$, $F(g(t))$ on $[a, b)$, take values in S and $f(a) = g(a)$, then $f = g$ on $[a, b]$; and the mirror statement with $f(b) = g(b)$ and left derivatives. (Mathlib: `ODE_solution_unique_of_mem_Icc_right`, `ODE_solution_unique_of_mem_Icc_left`.) Exists.

(E3) C^1 *implies locally Lipschitz.* If F is C^1 at p there are L and a neighbourhood on which F is L-Lipschitz (Mathlib: `ContDiffAt.exists_lipschitzOnWith`); and a function locally Lipschitz on a compact set is Lipschitz there (Mathlib: `LocallyLipschitzOn.exists_lipschitzOnWith_of_compact`, `Topology/Algebra/MetricSpace/Lipschitz.lean`), which packages Lemma 4.4 once local Lipschitzness on K is extracted from C^1 on W. Exists.

(E4) *Mean value inequality*: [K:`bp:p1:ext-mvt`]; for a differentiable map on a convex set with $\|DF\| \leq L$, F is L-Lipschitz there; for a function of one variable with derivative bounded by M on an interval, $\|\alpha(u) - \alpha(s)\| \leq M|u - s|$. Exists.

(E5) *Topology.* Heine–Borel in E (`ProperSpace`); continuous images of compact sets are compact; sequential compactness; for a *nonempty* closed set A, $x \mapsto \operatorname{dist}(x, A)$ is continuous and positive exactly on $E \setminus A$ (`Metric.infDist`, `IsClosed.notMem_iff_infDist_pos`, which requires `A.Nonempty`; for $A = \emptyset$ Mathlib has $\operatorname{infDist}(x, \emptyset) = 0$, `Metric.infDist_empty`, so no statement below uses infDist of an empty set: thickenings are defined by $K_\delta = \{y : \exists x \in K,\ \|y - x\| \leq \delta\}$, which is empty for $K = \emptyset$). Exists.

(E6) *Calculus.* Chain rule `HasFDerivAt.comp_hasDerivAt`; a function continuous on $[s,t]$ with right derivative 0 on $[s,t)$ is constant (`constant_of_has_deriv_right_zero`); two one-sided derivatives agree ⇒ two-sided one (`HasDerivWithinAt.union` with $(-\infty,b]\cup[b,\infty)=\mathbb{R}$ and `hasDerivWithinAt_univ`); `ContDiffOn.fderiv_of_isOpen`: if H is C^{k+1} on an open set then DH is C^k there. Exists.

(E7) *Uniform existence time near a point.* If F is C^1 at x_0 and $t_0\in\mathbb{R}$ there are $r,\varepsilon>0$ such that for every $x\in\overline{B}(x_0,r)$ there is α with $\alpha(t_0)=x$ and derivative $F(\alpha(t))$ for all $t\in(t_0-\varepsilon,t_0+\varepsilon)$ (Mathlib: `ContDiffAt.exists_forall_mem_closedBall_exists_eq_forall_mem_Ioo_hasDerivAt`, `ExistUnique.lean:145`; note the solution is for the total F and is not asserted to stay in the neighbourhood of x_0). Exists. It covers Theorem 4.10 for a single point after the staying-in-W step; the explicit $\delta/\max(M,1)$ time of Proposition 4.9, uniform over arbitrary sets Φ with uniform data, is derived from (E1).

Not in Mathlib (checked in `Analysis/ODE` and `Geometry/Manifold/IntegralCurve`: no maximal-solution, continuation or first-exit statement, and no uniqueness on arbitrary intervals with only local Lipschitz constants): the following must be proved from the blueprint below: uniform data on arbitrary sets and its compact and periodic instances (Lemmas 4.6, 4.7), Lemma 4.8, Proposition 4.9 (derived from (E1); the existing statement (E7) may be reused for the single-point case), Theorems 4.11, 4.13, 4.14, Lemma 4.19 and the Hamiltonian layer.

Remark 4.1 (Lean totality). `KAM.hamiltonianVectorField` H uses `fderiv`, which is 0 at a point of non-differentiability, and the Newtonian fields use $1/\ell^3$ with $\ell=$ `planeLength` or `radius`. In every use below these are guarded: X_H occurs only on an open set where H is C^2 (or inside the frozen predicates, which assert differentiability at every visited point), and $\ell>0$ at every visited point because the frozen predicates require it. The general layer never uses F off W.

4.2. **The general solution predicate; Lipschitz data.**

Definition 4.2 (Solutions in W)**.** For $F:E\to E$, $W\subset E$, $J\subset\mathbb{R}$ and $z:\mathbb{R}\to E$: `IsSolIn` $F\,W\,J\,z$ holds if for every $t\in J$, $z(t)\in W$ and z has derivative $F(z(t))$ at t (two-sided). Suggested Lean name `KAM.ODE.IsSolutionIn`; it is *not* a frozen declaration.
Source: Standard; shape dictated by the frozen predicates of Definition 1.16 and §4.8.

Elementary consequences (each a one-line Lean lemma, (E6)): (i) restriction: `IsSolIn` $F\,W\,J\,z$ and $J'\subset J$ give `IsSolIn` $F\,W\,J'\,z$; (ii) z is continuous on J (differentiable at each point of J); (iii) *local nature*: if J' is a neighbourhood of each of its points and $z=y$ on it, then `IsSolIn` $F\,W\,J'\,y\Leftrightarrow$ `IsSolIn` $F\,W\,J'\,z$ for open J'.

Lemma 4.3 (Symmetries)**.** *Let* `IsSolIn` $F\,W\,J\,z$.

(a) (time shift) For $c\in\mathbb{R}$, $t\mapsto z(t+c)$ is a solution on $J-c$.
(b) (time reversal) $t\mapsto z(-t)$ is a solution of $-F$ on $-J$, and $-F$ is C^1 on W if F is.
(c) (space translation) If $v\in E$ satisfies $W+v=W$ and $F(x+v)=F(x)$ for $x\in W$, then $t\mapsto z(t)+v$ is a solution of F in W on J.

Source: Standard.

Proof. **Step S4.1: Chain rule.** (a) and (b) are the chain rule for $t\mapsto t+c$, $t\mapsto -t$ ((E6)); (c): $z+v$ has the same derivative $F(z(t))=F(z(t)+v)$, and $z(t)+v\in W+v=W$. □

Lemma 4.4 (Bounds and Lipschitz constants on compact sets)**.** *Let F be C^1 on the open set W and $K\subset W$ compact. Then F is bounded on K and there is $L\geq 0$ with $\|F(x)-F(y)\|\leq L\|x-y\|$ for $x,y\in K$.*
Source: Standard.

Proof. **Step S4.2: Via Mathlib.** Alternatively: F is locally Lipschitz at every point of W ((E3)), hence locally Lipschitz on K, and `LocallyLipschitzOn.exists_lipschitzOnWith_of_compact` ((E3)) gives L. The remaining steps are the explanatory proof of that fact.

Step S4.3: Bounded. F is continuous on $W \supset K$, so $\|F\| \le M$ on the compact K ((E5)).

Step S4.4: Contradiction hypothesis. If no L works, for every $k \in \mathbb{N}$, $k \ge 1$, there are $x_k \ne y_k$ in K with $\|F(x_k) - F(y_k)\| > k\|x_k - y_k\|$; then $\|x_k - y_k\| < 2M/k \to 0$.

Step S4.5: Compactness. A subsequence $x_{k_j} \to p \in K$, hence also $y_{k_j} \to p$. By (E3) there are L_p and a neighbourhood N_p of p on which F is L_p-Lipschitz. For large j both points lie in N_p and $k_j > L_p$, which contradicts Step S4.4. □

Definition 4.5 (Uniform data)**.** Let F be C^1 on the open set W. A set $\Phi \subset E$ has *uniform data* (δ, M, L), with $\delta > 0$, $M, L \ge 0$, if for every $x \in \Phi$: $\overline{B}(x, \delta) \subset W$, $\|F\| \le M$ on $\overline{B}(x, \delta)$ and F is L-Lipschitz on $\overline{B}(x, \delta)$. (Balls are convex, so by (E4) a bound $\|DF\| \le L$ on $\overline{B}(x, \delta) \subset W$ gives the Lipschitz clause.)

Lemma 4.6 (Compact sets have uniform data)**.** *Let F be C^1 on the open set W and $K \subset W$ compact. Then K has uniform data (δ, M, L) for some $\delta > 0$, and $K_\delta = \{y : \exists x \in K,\ \|y - x\| \le \delta\} \subset W$ is compact.*

Proof. **Step S4.6: Choice of δ.** If $K = \emptyset$ put $\delta = 1$: then $K_\delta = \emptyset$ and all assertions are vacuous (no infDist of the empty set is used). If $W = E$ put $\delta = 1$. Otherwise $K \ne \emptyset$ and $A = E \setminus W$ is closed and nonempty, so $x \mapsto \operatorname{dist}(x, A)$ is continuous and positive on K ((E5)); it has a minimum $2\delta > 0$ on the compact nonempty K, and then $K_\delta \subset W$. K_δ is the image of the compact $K \times \overline{B}(0, \delta)$ under $(x, w) \mapsto x + w$, hence compact.

Step S4.7: Data. Apply Lemma 4.4 to K_δ: $\|F\| \le M$ and F is L-Lipschitz on $K_\delta \supset \overline{B}(x, \delta)$ for $x \in K$. □

Lemma 4.7 (Uniform data for angle-periodic fields)**.** *Let $E = \mathbb{R}^n \times \mathbb{R}^n$ (max norm), $U \subset \mathbb{R}^n$ open, $W = \mathbb{R}^n \times U$, F C^1 on W with $F(\theta + k, I) = F(\theta, I)$ for $(\theta, I) \in W$, $k \in \mathbb{Z}^n$. If $K \subset U$ is compact, then $\Phi = \mathbb{R}^n \times K$ has uniform data (δ, M, L) for some $\delta \in (0, 1]$.*

Proof. **Step S4.8: A compact cell.** If $K = \emptyset$ then $\Phi = \emptyset$ and any δ, M, L work. Otherwise choose $\delta \in (0, 1]$ with $K_\delta = \{I : \exists J \in K,\ |I - J| \le \delta\} \subset U$ (as in Step S4.6, using the nonempty closed set $\mathbb{R}^n \setminus U$, or $\delta = 1$ if $U = \mathbb{R}^n$). $C = [-1, 2]^n \times K_\delta$ is compact and contained in W; let M, L_C be the constants of Lemma 4.4 for C.

Step S4.9: Translating a ball into the cell. Let $x = (\theta, I) \in \Phi$, $k = (\lfloor \theta_i \rfloor)_i \in \mathbb{Z}^n$ and $x' = x - (k, 0) \in [0, 1)^n \times K$. Since the norm is the max norm and $\delta \le 1$, $\overline{B}(x', \delta) \subset C$, and $\overline{B}(x, \delta) = \overline{B}(x', \delta) + (k, 0) \subset W$. By periodicity $F(y + (k, 0)) = F(y)$ on $\overline{B}(x', \delta)$, so $\|F\| \le M$ and F is L_C-Lipschitz on $\overline{B}(x, \delta)$. □

4.3. **Local existence.**

Lemma 4.8 (A priori short-time confinement)**.** *Let J be an interval, $t_0 \in J$, $\alpha : \mathbb{R} \to E$ with $\alpha(t_0) = x$ and $\alpha'(t) = F(\alpha(t))$ for all $t \in J$ (derivative two-sided; F is only assumed to be a function), and suppose $\|F\| \le M$ on $\overline{B}(x, \delta)$ with $M > 0$. Then $\|\alpha(t) - x\| \le M|t - t_0|$ for all $t \in J$ with $|t - t_0| < \delta/M$; in particular $\alpha(t) \in \overline{B}(x, \delta)$.*
Source: Standard.

Proof. **Step S4.10: Forward bootstrap.** Let $t \in J$, $t_0 < t < t_0 + \delta/M$ (the case $t < t_0$ follows by Lemma 4.3(b) and the case $t = t_0$ is trivial). Let $A = \{s \in [t_0, t] : \|\alpha(u) - x\| \le M(u - t_0)\ \forall u \in [t_0, s]\}$ and $s^* = \sup A$. $t_0 \in A$ and A is closed by continuity of α on J, so $s^* \in A$.

Step S4.11: Extension. If $s^* < t$, then $\|\alpha(s^*) - x\| \le M(s^* - t_0) < \delta$, so by continuity there is $\eta > 0$, $s^* + \eta \le t$, with $\alpha([s^*, s^* + \eta]) \subset \overline{B}(x, \delta)$. On this interval $\|\alpha'\| = \|F(\alpha)\| \le M$, so by (E4) $\|\alpha(u) - \alpha(s^*)\| \le M(u - s^*)$ and hence $\|\alpha(u) - x\| \le M(u - t_0)$ there, contradicting the definition of s^*. So $s^* = t$. □

Proposition 4.9 (Local existence with uniform time)**.** *Let F be C^1 on the open set W and Φ have uniform data (δ, M, L); put $M' = \max(M, 1)$ and $\varepsilon = \delta/M'$. Then for every $x \in \Phi$ and every $t_0 \in \mathbb{R}$ there is $\alpha : \mathbb{R} \to E$ with $\alpha(t_0) = x$, $\|\alpha(t) - x\| < \delta$ for all $t \in (t_0 - \varepsilon, t_0 + \varepsilon)$ and $\mathtt{IsSolIn}\, F\, W\, (t_0 - \varepsilon, t_0 + \varepsilon)\, \alpha$.*
Source: Standard (Picard–Lindelöf).

Proof. **Step S4.12: Picard–Lindelöf.** Apply (E1) with $a = \delta$, bound M', Lipschitz constant L and $T = \varepsilon$ ($M'T = \delta \le a$): there is α with $\alpha(t_0) = x$ and derivative $F(\alpha(t))$ within $[t_0 - \varepsilon, t_0 + \varepsilon]$ at each point of it. At points t of the open interval this is a two-sided derivative, since the closed interval is a neighbourhood of t.

Step S4.13: Staying in W. By Lemma 4.8 on $J = (t_0 - \varepsilon, t_0 + \varepsilon)$, $\|\alpha(t) - x\| \le M'|t - t_0| < \delta$, so $\alpha(t) \in \overline{B}(x, \delta) \subset W$. □

Theorem 4.10 (Local existence, both time directions)**.** *Let F be C^1 on the open set W, $x_0 \in W$, $t_0 \in \mathbb{R}$. There are $\varepsilon > 0$ and $\alpha : \mathbb{R} \to E$ with $\alpha(t_0) = x_0$ and $\mathtt{IsSolIn}\, F\, W\, (t_0 - \varepsilon, t_0 + \varepsilon)\, \alpha$. Moreover ε may be taken uniform on any compact $K \subset W$ (for all $x_0 \in K$).*
Source: Standard.

Proof. **Step S4.14: Reduction.** $\{x_0\}$ (resp. K) is compact; Lemma 4.6 gives uniform data and Proposition 4.9 the claim. □

4.4. Uniqueness on every interval.

Theorem 4.11 (Uniqueness on arbitrary intervals)**.** *Let F be C^1 on the open set W, $J \subset \mathbb{R}$ an interval (of any kind: open, closed, half-open, unbounded, or a point), $t_0 \in J$, and $y_1, y_2 : \mathbb{R} \to E$ with $\mathtt{IsSolIn}\, F\, W\, J\, y_i$ and $y_1(t_0) = y_2(t_0)$. Then $y_1 = y_2$ on J. No confinement, boundedness or smallness of either solution is assumed. In particular, if y_1, y_2 are solutions on intervals J_1, J_2 and agree at one point of $J_1 \cap J_2$, they agree on $J_1 \cap J_2$ (an interval), and two solutions which agree at one time of J agree at all times of J.*
Source: Standard; used for BM11 Theorem 2.1 (uniqueness) and in the maximal-solution argument of BM11 p. 13.

Proof. **Step S4.15: A compact piece.** Let $t_1 \in J$, $t_1 > t_0$ ($t_1 < t_0$ is the mirror image, using the left-derivative form of (E2); $t_1 = t_0$ is the hypothesis). Then $[t_0, t_1] \subset J$, each y_i is continuous on it, so $K = y_1([t_0, t_1]) \cup y_2([t_0, t_1])$ is compact ((E5)) and $K \subset W$.

Step S4.16: Lipschitz on K. Lemma 4.4 gives L with F L-Lipschitz on K.

Step S4.17: Apply (E2). Both y_i are continuous on $[t_0, t_1]$, have right derivative $F(y_i(t))$ on $[t_0, t_1)$ (a two-sided derivative gives a right derivative) and take values in K; so $y_1(t_1) = y_2(t_1)$. □

4.5. Maximal solutions and continuation.

Lemma 4.12 (Extension to an open interval)**.** *Let F be C^1 on the open set W, $J \neq \emptyset$ an interval and $\mathtt{IsSolIn}\, F\, W\, J\, y$. Then there are an open interval $J' \supset J$ and y' with $y' = y$ on J and $\mathtt{IsSolIn}\, F\, W\, J'\, y'$.*

Proof. **Step S4.18: Right end.** If J has a largest element b, apply Theorem 4.10 at $x_0 = y(b) \in W$, $t_0 = b$: α on $(b-\varepsilon, b+\varepsilon)$. Put $y_1(s) = y(s)$ for $s \le b$ and $y_1(s) = \alpha(s)$ for $s > b$. Then y_1 is a solution on $J_1 = J \cup (b, b+\varepsilon)$: at $t \in J$, $t < b$, and at $t \in (b, b+\varepsilon)$ it agrees with y, resp. α, near t; at $t = b$ it has derivative $F(y(b))$ within $(-\infty, b]$ (it equals y there) and within $[b, \infty)$ (it equals α there, as $\alpha(b) = y(b)$), so by (E6) it has the two-sided derivative. If J has no largest element, $J_1 = J$, $y_1 = y$.

Step S4.19: Left end. Apply Step S4.18 to the reversed solution $y_1(-\cdot)$ (Lemma 4.3(b)). The result has no endpoint, so is an open interval. □

Theorem 4.13 (Maximal solution)**.** *Let F be C^1 on the open set W and $x_0 \in W$. There are $T_- \in [-\infty, 0)$, $T_+ \in (0, +\infty]$ and $z : \mathbb{R} \to E$ with $z(0) = x_0$ and* `IsSolIn` *$F\, W\, (T_-, T_+)\, z$, such that every solution y with $y(0) = x_0$ on an interval $J \ni 0$ satisfies $J \subset (T_-, T_+)$ and $y = z$ on J. Then (T_-, T_+) is the* maximal interval *and $z|_{(T_-,T_+)}$ the* maximal solution*; both are unique. For initial time t_0 instead of 0 use Lemma 4.3(a).*
Source: BM11 p. 13 ("maximal time of existence"); standard.

Proof. **Step S4.20: The union.** Let $\mathcal{J}$ be the set of open intervals $J \ni 0$ carrying a solution y_J with $y_J(0) = x_0$; $\mathcal{J} \neq \emptyset$ by Theorem 4.10. The union I of $\mathcal{J}$ is an open interval $(T_-, T_+) \ni 0$ (a union of intervals containing 0).

Step S4.21: Gluing. For $t \in I$ choose $J \in \mathcal{J}$ with $t \in J$ and set $z(t) = y_J(t)$ (and $z(t) = x_0$ outside I). This is independent of J: for $J, J' \in \mathcal{J}$, $y_J = y_{J'}$ on $J \cap J' \ni 0$ by Theorem 4.11. Each $t \in I$ has an open neighbourhood $J \in \mathcal{J}$ on which $z = y_J$, so z is a solution on I (local nature, (iii) above).

Step S4.22: Maximality. If y is a solution on an interval $J \ni 0$ with $y(0) = x_0$, Lemma 4.12 gives an open $J' \supset J$ and a solution y' extending y, so $J' \in \mathcal{J}$, $J \subset J' \subset I$, and $y = y' = z$ on J (Theorem 4.11). Uniqueness of the interval (T_-, T_+), and of z as a function on it (equality on this interval of any two representatives; values outside it are arbitrary), is immediate from the maximality property applied twice. □

Theorem 4.14 (Continuation criterion, uniform form)**.** *Let F be C^1 on the open set W, $x_0 \in W$, (T_-, T_+) and z the maximal interval and solution of Theorem 4.13. Let $\Phi \subset E$ have uniform data (δ, M, L) (Definition 4.5). If $T_+ < \infty$ there is no $t_* \in [0, T_+)$ with $z(t) \in \Phi$ for all $t \in [t_*, T_+)$. Symmetrically, if $T_- > -\infty$ there is no $t_* \in (T_-, 0]$ with $z(t) \in \Phi$ for all $t \in (T_-, t_*]$. Equivalently: if the solution stays in Φ near the right (left) end of its maximal interval, that end is $+\infty$ ($-\infty$).*
Source: Standard; BM11 p. 13 ("this solution is defined for all time"), Pöschel 1993 p. 198 ("$T = T_*$").

Proof. **Step S4.23: Set-up.** Suppose $T_+ < \infty$ and $z(t) \in \Phi$ on $[t_*, T_+)$. Let $\varepsilon = \delta / \max(M, 1)$ from Proposition 4.9. Choose $t_1 \in [t_*, T_+)$ with $t_1 > T_+ - \varepsilon$.

Step S4.24: A solution through $z(t_1)$. Proposition 4.9 with $x = z(t_1) \in \Phi$ and $t_0 = t_1$ gives α on $(t_1 - \varepsilon, t_1 + \varepsilon)$ with $\alpha(t_1) = z(t_1)$. By Theorem 4.11 on $(t_1 - \varepsilon, t_1 + \varepsilon) \cap (T_-, T_+)$, $\alpha = z$ there.

Step S4.25: Gluing past T_+. Put $y = z$ on (T_-, T_+) and $y = \alpha$ on $[T_+, t_1 + \varepsilon)$; then $y = \alpha$ on $(t_1 - \varepsilon, t_1 + \varepsilon)$ and $y = z$ on (T_-, T_+), two open sets covering $(T_-, t_1 + \varepsilon)$ on each of which y is a solution. So y is a solution on $(T_-, t_1 + \varepsilon) \ni 0$ with $y(0) = x_0$, and $t_1 + \varepsilon > T_+$ contradicts Theorem 4.13. The left end follows by Lemma 4.3(b). □

Corollary 4.15 (Compact continuation)**.** *In the setting of Theorem 4.14, if $K \subset W$ is compact and $z(t) \in K$ for all $t \in [t_*, T_+)$ (some $t_* \in [0, T_+)$), then $T_+ = +\infty$; likewise on the left.*

Proof. Lemma 4.6 gives uniform data for K; apply Theorem 4.14. □

Corollary 4.16 (Closed sets, periodic fields)**.** *Let $E = \mathbb{R}^n \times \mathbb{R}^n$, $U \subset \mathbb{R}^n$ open, $W = \mathbb{R}^n \times U$, F C^1 on W with $F(\theta + k, I) = F(\theta, I)$ for $k \in \mathbb{Z}^n$. If the maximal solution through $x_0 \in W$ satisfies $z(t)_2 \in K$ for $t \in [t_*, T_+)$ with $K \subset U$ compact, then $T_+ = +\infty$ (likewise on the left). The angle components $z(t)_1$ may be unbounded.*

Proof. Lemma 4.7 gives uniform data for $\Phi = \mathbb{R}^n \times K$; apply Theorem 4.14. □

Remark 4.17 (What the closed-set variant really needs). Staying in a closed non-compact set $\Phi \subset W$ does *not* suffice in general (the solution of $\dot{x} = x^2$ on $E = \mathbb{R}$, $W = \mathbb{R}$, $x_0 = 1$, $x(t) = 1/(1-t)$, stays in the closed set $[1, \infty)$ for $0 \le t < 1$ and blows up at $T_+ = 1$). What suffices is the uniform data (δ, M, L) of Definition 4.5: a δ-neighbourhood of Φ inside W on which F is bounded and Lipschitz, uniformly. This holds for compact sets, for angle-periodic fields over a compact action set (Lemma 4.7), and for the Newtonian regions bounded away from collisions with bounded velocities (Corollary 4.27).

Theorem 4.18 (Confinement glue)**.** *Let F be C^1 on the open set W, $x_0 \in W$, $\tau > 0$, and $\Phi \subset E$ a set with uniform data (e.g. compact $\Phi \subset W$, or as in Lemma 4.7). Assume the* a priori confinement*: every solution y on an open interval $J \ni 0$ with $y(0) = x_0$ satisfies $y(t) \in \Phi$ for all $t \in J$ with $|t| \le \tau$. Then the maximal interval contains $[-\tau, \tau]$ in its interior, so z is a solution on $[-\tau, \tau]$ with $z(t) \in \Phi$ for $|t| \le \tau$, and every solution on $[-L, L]$ $(0 \le L \le \tau)$ with initial value x_0 equals z there.*
Source: BM11 p. 13 (the two-case argument "Therefore this solution is defined for all time"); Pöschel 1993 p. 198.

Proof. **Step S4.26: No exit before τ.** Suppose $T_+ \le \tau$. The maximal solution z on (T_-, T_+) satisfies, for $t \in [0, T_+)$ $(|t| < \tau)$, $z(t) \in \Phi$ by confinement. Theorem 4.14 gives $T_+ = \infty$, a contradiction. So $T_+ > \tau$, and likewise $T_- > -\tau$.

Step S4.27: Conclusion. $[-\tau, \tau] \subset (T_-, T_+)$ so z is a solution on $[-\tau, \tau]$ and $z(t) \in \Phi$ for $|t| \le \tau$ by confinement applied to z itself. Uniqueness is Theorem 4.11. □

4.6. **First exit from an open set; barriers.**

Lemma 4.19 (First exit)**.** *Let* `IsSolIn` $F\, W\, J\, z$ *with J an interval, $0 \in J$, $V \subset E$ open and $z(0) \in V$. Put $A = \{t \in J : t \ge 0,\ z(t) \notin V\}$. Then exactly one of the following holds.*

(i) $A = \emptyset$: $z(t) \in V$ for all $t \in J$, $t \ge 0$.

(ii) $A \neq \emptyset$. Then $t^* = \min A$ exists, $t^* > 0$, $t^* \in J$, $z(t) \in V$ for $t \in [0, t^*)$ and $z(t^*) \in \partial V = \overline{V} \setminus V$. The number t^* is the first exit time *and equals $\sup\{t \in J : t \ge 0,\ z([0,t]) \subset V\}$; in particular t^* is finite and lies in J (t^* is "inside the solution's interval" and the solution is defined, continuous, at t^*).*

The same holds backwards in time (replace $z(t)$ by $z(-t)$, Lemma 4.3(b)). Only continuity of z on J is used; F and W play no role.
Source: Standard; used for BM11 p. 13 ("smallest time t_* such that $\omega(t_*) \in R_K$") and Pöschel 1993 p. 198–199 (stopping-time arguments).

Proof. **Step S4.28: Closedness.** z is continuous on J (consequence (ii)), $E \setminus V$ is closed, so A is closed in $J \cap [0, \infty)$.

Step S4.29: The minimum. If $A \neq \emptyset$, $t^* = \inf A \in [0, \infty)$; $t^* \in J$ since J is an interval containing 0 and a point of A that is $\ge t^*$. By closedness $t^* \in A$, i.e. $z(t^*) \notin V$. As $z(0) \in V$, $t^* > 0$.

Step S4.30: Boundary point. For $t \in [0, t^*)$, $t \notin A$, so $z(t) \in V$. By continuity $z(t^*) = \lim_{t \uparrow t^*} z(t) \in \overline{V}$, and $z(t^*) \notin V$; as V is open, $\overline{V} \setminus V = \partial V$.

Step S4.31: The supremum formula. $z([0,t]) \subset V$ iff $t < t^*$ (for $t \in J$, $t \ge 0$), so the supremum is t^*. □

Definition 4.20 (Invariant sets)**.** $S \subset E$ is *invariant* for (F, W) if for every interval J' and every y with `IsSolIn` $F\,W\,J'\,y$: if $y(t_1) \in S$ for one $t_1 \in J'$ then $y(t) \in S$ for all $t \in J'$. (Equilibria $\{x : F(x) = 0\}$ are invariant by Theorem 4.11; so is any union of such sets and any set $\{G = c\}$ for a first integral G.)

Corollary 4.21 (Barrier)**.** *Let F be C^1 on the open set W, V open, `IsSolIn` $F\,W\,J\,z$ with J an interval, $0 \in J$, $z(0) \in V$. Let $G \subset E$ with $z(J) \subset G$, and suppose $\partial V \cap G \subset S$ where S is invariant and $z(0) \notin S$ (e.g. G an energy level, Theorem 4.23, and $\partial V \cap G$ a union of invariant sets not meeting the initial point). Then $z(t) \in V$ for all $t \in J$ (both time directions).*

Proof. **Step S4.32: Forward.** If not, Lemma 4.19(ii) gives $t^* \in J$ with $z(t^*) \in \partial V \cap G \subset S$; invariance gives $z(0) \in S$, a contradiction.

Step S4.33: Backward. Apply Step S4.32 to $t \mapsto z(-t)$, a solution of $-F$ on $-J$; G, ∂V are unchanged and S is invariant for $-F$ as well (reversal maps solutions of F to solutions of $-F$ and back). □

4.7. **Hamiltonian layer.** Phase space $P = \mathbb{R}^n \times \mathbb{R}^n$, points $z = (\theta, I)$, max norm. X_H is the frozen `KAM.hamiltonianVectorField`, Definition 1.3 (convention of [K:bp:p0:hamiltonian]): $X_H(z) = \big((\partial_{I_i} H(z))_i, (-\partial_{\theta_i} H(z))_i\big)$, with $\partial_{I_i} H(z) = DH(z)(0, e_i)$ and $\partial_{\theta_i} H(z) = DH(z)(e_i, 0)$.

Lemma 4.22 (Regularity of X_H)**.** *Let $W \subset P$ be open and $H : P \to \mathbb{R}$ be C^{k+1} on W, $k \geq 0$. Then X_H is C^k on W; for $k \geq 1$ it is C^1 (hence locally Lipschitz) on W, and H is differentiable at every point of W.*
Source: Standard.

Proof. **Step S4.34: Derivative is C^k.** $DH : W \to \mathcal{L}(P, \mathbb{R})$ is C^k on the open W ((E6), `ContDiffOn.fderiv_of_isOpen`).

Step S4.35: Components. Each coordinate of X_H is $\pm$ the composition of DH with the continuous linear evaluation $\ell \mapsto \ell(0, e_i)$ or $\ell \mapsto \ell(e_i, 0)$; hence C^k, and a map into $\mathbb{R}^n \times \mathbb{R}^n$ with C^k coordinates is C^k. □

Theorem 4.23 (Energy conservation)**.** *Let $H : P \to \mathbb{R}$, $J \subset \mathbb{R}$ an interval and $z : \mathbb{R} \to P$ such that for every $t \in J$, H is differentiable at $z(t)$ and z has derivative $X_H(z(t))$ at t. Then $t \mapsto H(z(t))$ is constant on J. (No regularity of H beyond the stated pointwise differentiability, and no confinement.)*
Source: BM11 p. 13 and Pöschel 1993 p. 198 ("energy conservation", "$\Delta H = H(I(t), \theta(t))|_0^T$ vanishes"); standard.

Proof. **Step S4.36: Chain rule.** For $t \in J$, $\frac{d}{dt} H(z(t)) = DH(z(t))\big[X_H(z(t))\big]$ ((E6)).

Step S4.37: Cancellation. Write $z = z(t)$, $\ell = DH(z)$ and $X_H(z) = (a, b)$ with $a_i = \ell(0, e_i)$, $b_i = -\ell(e_i, 0)$. By linearity, $\ell(a, b) = \sum_i a_i\, \ell(e_i, 0) + \sum_i b_i\, \ell(0, e_i) = \sum_i \ell(0, e_i)\ell(e_i, 0) - \sum_i \ell(e_i, 0)\ell(0, e_i) = 0$.

Step S4.38: Constancy. Let $s < t$ in J. $H \circ z$ is continuous on $[s, t] \subset J$ and has derivative 0 at every point, so $H(z(t)) = H(z(s))$ ((E6)). □

Lemma 4.24 (Periodicity in the angles passes to solutions)**.** *Let $W \subset P$ be open with $W + (k, 0) = W$ for all $k \in \mathbb{Z}^n$, and $H : P \to \mathbb{R}$ with $H(\theta + k, I) = H(\theta, I)$ for $(\theta, I) \in W$, $k \in \mathbb{Z}^n$. Then $X_H(\theta + k, I) = X_H(\theta, I)$ on W. Consequently, if `IsSolIn` $X_H\,W\,J\,z$ then `IsSolIn` $X_H\,W\,J\,(z + (k, 0))$ for every $k \in \mathbb{Z}^n$, and $H \circ (z + (k, 0)) = H \circ z$.*
Source: Standard.

Proof. **Step S4.39: Field.** Let $\tau_k(x) = x + (k, 0)$. On the open set W, $H \circ \tau_k = H$, so $DH = D(H \circ \tau_k) = DH \circ \tau_k$ at each point of W (the derivative of a translate; both sides are 0 if H is not differentiable, by the definition of `fderiv`). Evaluating on $(0, e_i)$ and $(e_i, 0)$ gives $X_H \circ \tau_k = X_H$ on W.

Step S4.40: Solutions. Lemma 4.3(c) with $v = (k, 0)$ applies to $F = X_H$. □

4.8. **The frozen solution predicates are instances.**

Proposition 4.25 (Track N and track I Hamiltonian predicates)**.** *Let $H : P \to \mathbb{R}$, $U \subset \mathbb{R}^n$ open, $W \subset P$ open with $\mathbb{R}^n \times U \subset W$, and H be C^2 on W. Put $W_U = \mathbb{R}^n \times U$. For every $J \subset \mathbb{R}$ and $z : \mathbb{R} \to P$,*

$$\texttt{IsSolutionOn}\, H\, U\, J\, z \iff \texttt{IsSolIn}\, X_H\, W_U\, J\, z \iff \texttt{IsHamiltonianSolutionOn}\, H\, U\, J\, z,$$

and X_H is C^1 on W_U. (The two frozen Hamiltonian predicates have the same formula up to the letter naming the action set, and under the regularity hypotheses both are equivalent to `IsSolIn` as shown below, `KAM.Nekhoroshev.IsSolutionOn` and `KAM.Iso.IsHamiltonianSolutionOn`.)
Frozen Lean declaration: `KAM.Nekhoroshev.IsSolutionOn`, `KAM.Iso.IsHamiltonianSolutionOn`
Source: Ours (bookkeeping).

Proof. **Step S4.41: Membership.** $z(t)_2 \in U \iff z(t) \in W_U$ by definition of W_U.

Step S4.42: Differentiability is free. For $z(t) \in W_U \subset W$, H is C^2 on the open W, hence differentiable at $z(t)$ (Lemma 4.22); so the clause `DifferentiableAt` $\mathbb{R}\, H\, (z\, t)$ is implied by $z(t) \in W_U$, and the two predicates coincide. Smoothness of X_H: Lemma 4.22, $k = 1$, restricted to $W_U \subset W$. □

Proposition 4.26 (Newtonian predicates)**.** *Let $N \geq 2$, $G \in \mathbb{R}$, `mass` : $Fin\, N \to \mathbb{R}$. Let $E_N = (Fin\, N \to \mathbb{R}^2) \times (Fin\, N \to \mathbb{R}^2)$ (max norm), $\ell =$ `planeLength`,*

$$W_N = \{(q, v) : \ell(q_j - q_i) > 0 \ \forall i \neq j\}, \qquad F_N(q, v) = \Big(v,\ \big(\textstyle\sum_{j \neq i} \frac{G\, \mathrm{mass}_j}{\ell(q_j - q_i)^3}(q_j - q_i)\big)_i\Big).$$

Then W_N is open, F_N is C^∞ on W_N, and for $Q, V : \mathbb{R} \to (Fin\, N \to \mathbb{R}^2)$ with $x(t) = (Q(t), V(t))$: `IsNewtonSolution` $G\, \mathrm{mass}\, Q\, V \iff$ `IsSolIn` $F_N\, W_N\, \mathbb{R}\, x$. Likewise for `IsFixedSolution` $G\, m_0\, m\, \varepsilon$ with $W_N' = W_N \cap \{\ell(q_i) > 0 \ \forall i\}$ and the field $F'(q, v) = (v, (-\frac{Gm_0}{\ell(q_i)^3} q_i + \varepsilon \sum_{j \neq i} \frac{Gm_j}{\ell(q_j - q_i)^3}(q_j - q_i))_i)$, and for `IsRestrictedSolutionOn` $\mu\, J$ with $W_\mu = \{(q, p) :$ `CollisionFree` $\mu\, (q, p)\}$ and the total extension F_μ of `restrictedVectorField` μ by the same formula (with the values off W_μ irrelevant): `IsRestrictedSolutionOn` $\mu\, J\, z \iff$ `IsSolIn` $F_\mu\, W_\mu\, J\, z$. In all three cases F is C^∞ on the open set W and the general layer applies with $J = \mathbb{R}$ (resp. the given J).
Frozen Lean declaration: `KAM.Planetary.IsNewtonSolution`, `KAM.Planetary.IsFixedSolution`, `KAM.Iso.IsRestrictedSolutionOn`
Source: Ours (bookkeeping); Fejoz Section 6.1, Newton equations as in the frozen text.

Proof. **Step S4.43: Open domain and smoothness.** $\ell(w)^2 =$ `planeSq`$(w) = w_0^2 + w_1^2$ is a polynomial, so $\{\ell > 0\} = \{$`planeSq` $> 0\}$ is open, and $\ell = \sqrt{\cdot}$ is C^∞ where `planeSq` > 0. Hence $\ell(q_j - q_i)^{-3}$ is C^∞ on W_N and F_N is a finite sum of products of C^∞ maps. Same for F' and F_μ (`radius` $= \sqrt{\texttt{radiusSq}}$ with the shifts by $\texttt{primary}_{1,2}\mu$).

Step S4.44: Derivatives of pairs. A map into a finite product has derivative (a, b) at t iff each component has the corresponding derivative (`hasDerivAt_pi`, `HasDerivAt.prodMk` and converse). The first block of the frozen definition (distance positivity) is membership in W_N; the second block says Q has derivative $V(t)$; the third says V has derivative the force. Together they say exactly that x has derivative $F_N(x(t))$ at t. □

Corollary 4.27 (Newtonian continuation)**.** *In the setting of Proposition 4.26, let x be the maximal solution through $x_0 \in W_N$ (Theorem 4.13), $\rho > 0$, $\bar{V} \geq 0$ and*

$$\Phi_{\rho,\bar{V}} = \{(q,v) : \ \ell(q_j - q_i) \geq \rho \ (i \neq j), \ |v| \leq \bar{V}\}.$$

If $T_+ < \infty$ then for every $\rho, \bar{V}$ and every $t_* \in [0, T_+)$ there is $t \in [t_*, T_+)$ with $x(t) \notin \Phi_{\rho,\bar{V}}$ (a finite-time singularity needs near-collisions or unbounded velocities; positions may be unbounded). The same for F' with the additional condition $\ell(q_i) \geq \rho$, and for F_μ with $\ell \geq \rho$ from both primaries and $|q|, |p| \leq \bar{V}$.

Proof. **Step S4.45: Uniform data.** Let $\delta = \rho/8$. If $|y - x| \leq \delta$ (max norm) with $x \in \Phi_{\rho,\bar{V}}$ then each $\ell(q'_j - q'_i) \geq \rho - 2\sqrt{2}\,\delta \geq \rho/2$, so $y \in W_N$ and $\overline{B}(x,\delta) \subset W_N$. On this ball $\ell \geq \rho/2$ and $|v'| \leq \bar{V} + \delta$; the entries of F_N and DF_N are polynomials in v', the differences $q'_j - q'_i$ (Euclidean length $\leq \ell(q'_j - q'_i)$) and the inverse powers ℓ^{-3}, ℓ^{-5} ($D(w/\ell^3) = \ell^{-3}I - 3\ell^{-5}ww^\top$), hence bounded by a constant $C(N, G, \mathrm{mass}, \rho, \bar{V})$ *independent of the position of x*. By (E4) on the convex ball, F_N is C-Lipschitz there. So $\Phi_{\rho,\bar{V}}$ has uniform data.

Step S4.46: Apply. Theorem 4.14 gives the claim; the other two fields are identical in structure (for F_μ the extra terms are linear in (q,p) and bounded on $|q|, |p| \leq \bar{V}$). □

4.9. **The track-N statements.** Throughout, $R, s > 0$, $h : \mathbb{C}^n \to \mathbb{C}$ with `IsAnalyticIntegrable` $R\,s\,h$, $\varepsilon \geq 0$ and $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ with `IsAnalyticPerturbation` $R\,s\,\varepsilon\,f$ (Definitions 1.11, 1.12), $H = H_{h,f}$ (Definition 1.4), $W_s = \mathbb{R}^n \times B(0, R+s) \subset P$ (open; $B(0,R+s) = \mathcal{B}_s(R)$ by Remark 1.9), and $U \subset B(0,R+s)$ open (in N-HEAD-C, $U = B(0,R)$).

Lemma 4.28 (Input from Part 1: H is smooth on W_s)**.** *$H_{h,f}$ is C^∞ (in particular C^2) on the open set W_s, $X_H = X_{h+f}$ on $W_s \cap (\mathbb{R}^n \times \mathbb{R}^n)$, and $H(\theta + k, I) = H(\theta, I)$ for $(\theta, I) \in W_s$, $k \in \mathbb{Z}^n$.*
Source: Part 1, Lemma 2.3(b)–(d) (real smoothness on W_s, field $X_H = X_{h+f} \circ \iota$, periodicity); ingredients [K:`bp:p1:holomorphic-regularity`], [K:`bp:p1:hamiltonian-real`].

Proof. **Step S4.47: Citation.** (b), (c), (d) of Lemma 2.3 are exactly the three claims. For orientation, the argument is as follows.

Step S4.48: Regularity. The function $G(\theta, I) = h(I) + f(\theta, I)$ is holomorphic on the open set $D_s(R) \supset W_s$ (real points (θ, I) with $I \in \mathcal{B}_s(R) = B(0,R)_s \cap \mathbb{R}^n$ lie in $T_s \times B(0,R)_s$), hence C^∞ in the real sense ([K:`bp:p1:holomorphic-regularity`]); it is real on W_s by Definition 1.11(iv) and 1.12(iii). So $H = \mathrm{Re}\, G = G$ on W_s is the restriction of C^∞ map to a real subspace: C^∞ on W_s; $X_H = X_G$ at real points is [K:`bp:p1:hamiltonian-real`]. (Packaged as Lemma 2.3.)

Step S4.49: Periodicity. $f(\theta + k, I) = f(\theta, I)$ for real θ, $I \in \mathcal{B}_s(R)$, and h depends on I only. □

Theorem 4.29 (X_H is C^1, hence locally Lipschitz)**.** *$X_{H_{h,f}}$ is C^1 on $W_s \supset \mathbb{R}^n \times B(0,R)$; consequently it is Lipschitz on every compact subset of W_s, and on every set $\mathbb{R}^n \times K$ with $K \subset B(0, R+s)$ compact there are δ, M, L as in Lemma 4.7.*
Source: BM11 p. 13 (implicit in "the solution"); standard.

Proof. **Step S4.50: Smoothness.** Lemma 4.28 and Lemma 4.22 ($k = 1$).

Step S4.51: Lipschitz. Lemma 4.4 on compact sets, Lemma 4.7 (with $U = B(0, R+s)$, using the periodicity of X_H from Lemma 4.24 and Lemma 4.28) for the cylinders. □

Theorem 4.30 (Local existence and uniqueness for `IsSolutionOn`)**.** *For every $t_0 \in \mathbb{R}$ and $z_0 \in \mathbb{R}^n \times U$ there are $\epsilon > 0$ and $y : \mathbb{R} \to P$ with $y(t_0) = z_0$ and* `IsSolutionOn` $H\,U\,(t_0 - \epsilon, t_0 + \epsilon)\,y$ *(both time directions). If J is any interval, $t_0 \in J$ and y_1, y_2 satisfy* `IsSolutionOn` $H\,U\,J\,y_i$ *with $y_1(t_0) = y_2(t_0)$, then $y_1 = y_2$ on J.*
Source: BM11 p. 13 and Theorem 2.1 p. 5; Picard–Lindelöf (external fact (E1)).

Proof. **Step S4.52: Reduction.** Proposition 4.25 (with $W = W_s$, H C^2 there by Lemma 4.28) turns $\texttt{IsSolutionOn}\, H\, U$ into $\texttt{IsSolIn}\, X_H\, (\mathbb{R}^n \times U)$ with X_H C^1 on $\mathbb{R}^n \times U$.

Step S4.53: Existence, uniqueness. Theorem 4.10 and Theorem 4.11. □

Theorem 4.31 (Energy conservation for $\texttt{IsSolutionOn}$)**.** *If $\texttt{IsSolutionOn}\, H\, U\, J\, z$ for an interval J (here H may be any function $P \to \mathbb{R}$ and U any set), then $H(z(t)) = H(z(t'))$ for $t, t' \in J$. In particular, for $H = H_{h,f}$ and $J = [0, T]$ the quantity $\Delta H = H(z(t))|_0^T$ of Pöschel's Lemma 1 vanishes.*
Source: Pöschel 1993 p. 198 ("by energy conservation ... ΔH vanishes"; page read); BM11 p. 13.

Proof. **Step S4.54: Direct.** The definition gives differentiability of H at every $z(t)$ and $z'(t) = X_H(z(t))$; Theorem 4.23. □

Theorem 4.32 (Maximal solutions and continuation)**.** *Let $z_0 = (\theta_0, I_0)$ with $I_0 \in U$. There is a unique maximal interval $(T_-, T_+) \ni 0$ and a solution z with $z(0) = z_0$ and $\texttt{IsSolutionOn}\, H\, U\, (T_-, T_+)\, z$, such that every y with $y(0) = z_0$ and $\texttt{IsSolutionOn}\, H\, U\, J\, y$ on an interval $J \ni 0$ has $J \subset (T_-, T_+)$ and $y = z$ on J (Theorem 4.13 through Proposition 4.25). If $K \subset U$ is compact and $z(t)_2 \in K$ for all $t \in [t_*, T_+)$ for some $t_* \in [0, T_+)$, then $T_+ = +\infty$; likewise on the left. The angle components $z(t)_1$ may be unbounded: no obstruction arises because X_H is 1-periodic in θ.*
Source: BM11 p. 13 ("From this estimate we can deduce that the solution $I(t)$ belongs to some compact ball ... this solution is defined for all time"); Pöschel 1993 p. 198 ("$T = T_*$").

Proof. **Step S4.55: General statements.** By Proposition 4.25, $\texttt{IsSolutionOn}\, H\, U\, J\, y \iff \texttt{IsSolIn}\, X_H\, (\mathbb{R}^n \times U)\, J\, y$ and X_H is C^1 on $W = \mathbb{R}^n \times U$; maximality is Theorem 4.13.

Step S4.56: Periodic cylinder. X_H is 1-periodic in θ on W (Lemmas 4.24, 4.28). Corollary 4.16 yields $T_+ = \infty$. □

Theorem 4.33 (Uniqueness on closed intervals)**.** *For $L \geq 0$ and y_1, y_2 with $y_1(0) = y_2(0)$ and $\texttt{IsSolutionOn}\, H\, U\, [-L, L]\, y_i$ one has $y_1 = y_2$ on $[-L, L]$. In particular a solution on $[-L, L]$ agrees with every solution on a larger interval containing $[-L, L]$ and the same initial value (clause (d) of Theorem 1.19, where $U = B(0, R)$, $L \in [0, \tau]$, y arbitrary and z the exhibited solution). No confinement of y is assumed: Gronwall is replaced by Theorem 4.11, whose Lipschitz constant comes from the compact set $y_1([-L, L]) \cup y_2([-L, L])$.*
Source: Standard (Gronwall); clause (d) of the frozen N-HEAD-C.

Proof. **Step S4.57: Apply.** $J = [-L, L]$, $t_0 = 0$ in Theorem 4.30 (uniqueness part). □

Theorem 4.34 (Confinement glue)**.** *Let $\theta_0 \in \mathbb{R}^n$, $I_0 \in U$, $\tau > 0$ and $\rho > 0$ with $\overline{B}(I_0, \rho) \subset U$ (e.g. $|I_0| < R/2$, $\rho \leq R/4$, $U = B(0, R)$). Assume that for every open interval $J \ni 0$ and every y with $y(0) = (\theta_0, I_0)$ and $\texttt{IsSolutionOn}\, H\, U\, J\, y$ one has $|y(t)_2 - I_0| \leq \rho$ for all $t \in J$ with $|t| \leq \tau$. Then there is z with $z(0) = (\theta_0, I_0)$, $\texttt{IsSolutionOn}\, H\, U\, [-\tau, \tau]\, z$ and $|z(t)_2 - I_0| \leq \rho$ for $|t| \leq \tau$; and for every $L \in [0, \tau]$ and every y with $y(0) = z(0)$ and $\texttt{IsSolutionOn}\, H\, U\, [-L, L]\, y$, $y = z$ on $[-L, L]$. This is the assembly of clauses (a)–(d) of Theorem 1.19 from an a priori estimate (Part 10); $\rho = C\varepsilon^{1/(2n)}$ there.*
Source: BM11 p. 13 (two cases, "Therefore this solution is defined for all time"); Pöschel 1993 p. 198.

Proof. **Step S4.58: Compact confinement set.** $K = \overline{B}(I_0, \rho)$ is compact and $\subset U$; $\Phi = \mathbb{R}^n \times K$ has uniform data by Lemma 4.7 (with $W = \mathbb{R}^n \times U$, $F = X_H$ periodic by Lemma 4.24).

Step S4.59: General glue. By Proposition 4.25 the hypothesis reads: every general solution on an open $J \ni 0$ with this initial value stays in Φ for $|t| \leq \tau$. Theorem 4.18 gives z on $[-\tau, \tau]$ in Φ, i.e. $|z(t)_2 - I_0| \leq \rho$, and uniqueness on $[-L, L]$ (it is also Theorem 4.33). □

4.10. **Label index for tracks I and P.**

`bpN:p3:gen-sol`, `gen-symmetry`	predicate; time shift, reversal, space translation
`gen-lip`, `gen-uniform`, `gen-uniform-compact`, `gen-uniform-periodic`	Lipschitz and uniform data
`gen-short`, `gen-uniform-exist`, `gen-local`	local existence, uniform time
`gen-uniqueness`	uniqueness on all intervals, no confinement
`gen-extend`, `gen-maximal`	maximal solutions
`gen-continuation`, `-compact`, `-periodic`, `gen-confinement`	continuation, closed sets with uniform bounds, confinement glue
`gen-exit`, `gen-invariant`, `gen-barrier`	first exit, barrier
`ham-smooth`, `ham-energy`, `ham-periodic`	Hamiltonian layer
`ham-reduction`, `newton-reduction`, `newton-continuation`	frozen predicates of I, N, P
`holo-input`, `lipschitz`, `local-exist`, `energy`, `continuation`, `uniqueness-closed`, `confinement-glue`	track N specializations

5. Weighted Fourier norms and Cauchy estimates (shared by tracks N, P, K)

This part is Appendix B and Appendix C of Pöschel 1993 (pp. 213–215) in the frozen conventions. It is shared infrastructure: track P (averaging, two-scale KAM) cites its results as [N:`bpN:p4:...`]. The statements are therefore made for functions $u(\theta, I)$ with $\theta \in \mathbb{T}^n$ and I in a neighbourhood of an *arbitrary* set $D \subset \mathbb{C}^m$, with m independent of n; only the Poisson bracket (Lemma 5.7) needs $m = n$. All norms on $\mathbb{R}^N$, $\mathbb{C}^N$ are max norms; no Euclidean norm occurs in this part.
Normalisation. Angles have period 1, characters are $\chi_k(\theta) = e^{2\pi i k\cdot\theta}$, and the strip width s is the *geometric* width, $T_s = \{\theta \in \mathbb{C}^n : |\operatorname{Im}\theta|_\infty < s\}$ ([K:`bp:p0:domains`]); the weight is $e^{2\pi|k|_1 s}$, that is, the coefficient radius is $2\pi s$ as in `docs/fourier-conventions.md`. With this choice every formula of Pöschel's Appendix B holds *verbatim*, with no factor 2π (Remark 5.10); for the θ-derivative the factor 2π of $\partial_{\theta_j}\chi_k = 2\pi i k_j \chi_k$ is absorbed by the weight. Widths are r (actions) and s (angles); the *losses* are ρ (actions) and σ (angles), as in Pöschel's (ρ, σ). (The brief's $|u|_{D,r,\sigma}$ is $|u|_{D,r,s}$ here.)

5.1. **External facts.** Besides [K:`bp:p1:ext-cauchy`], [K:`bp:p1:ext-integral`], [K:`bp:p1:ext-fourier`], [K:`bp:p1:ext-mvt`], the proofs use:

(E1) *Holomorphic dependence on a parameter.* Let $Q \subset \mathbb{R}^p$ be a compact box, $U \subset \mathbb{C}^q$ open, $G : Q \times U \to \mathbb{C}^d$ continuous, holomorphic in $I \in U$ for each $x \in Q$, with $D_I G$ continuous on $Q \times U$. Then $g(I) = \int_Q G(x, I)\,dx$ is holomorphic on U and $Dg(I) = \int_Q D_I G(x, I)\,dx$. (Mathlib: `hasFDerivAt_integral_of_dominated_of_fderiv_le` with a bound on a compact polydisc.) Exists.

(E2) *Uniform limits.* If K_j are holomorphic on an open $U \subset \mathbb{C}^q$ and $K_j \to K$ uniformly on U, then K is holomorphic on U, and $DK_j \to DK$ uniformly on $\{z : \bar{P}(z, \eta) \subset U\}$ for each $\eta > 0$ (the derivatives by Lemma 5.8 applied to $K_j - K_l$; holomorphy by [K:`bp:p1:uniform-limits`]). Exists in this form modulo that lemma.

(E3) *Series and integrals.* Tonelli for nonnegative double series and for integrals of nonnegative functions; the Cauchy product of two absolutely convergent series; termwise integration of a uniformly convergent series over a box ([K:`bp:p1:ext-integral`]). Exists.

5.2. **Domains, function class, norm.**

Definition 5.1 (Domains)**.** Let $m, n \geq 0$, $D \subset \mathbb{C}^m$ any subset, $r, s > 0$. Put $V_r D = \{I \in \mathbb{C}^m : \exists I' \in D,\ |I - I'|_\infty < r\}$ and $\mathcal{D}_{r,s} = T_s \times V_r D \subset \mathbb{C}^n \times \mathbb{C}^m$ (Pöschel's $V_{r,s}$). For $z = (\theta, I)$ let

$\bar P(z;a,b)=\{(\theta',I'):|\theta'-\theta|_\infty\le a,|I'-I|_\infty\le b\}$. Then: (a) V_rD and $\mathcal D_{r,s}$ are open, and $\mathcal D_{r',s'}\subset\mathcal D_{r,s}$ if $r'\le r$, $s'\le s$. (b) If $0<r'<r$, $0<s'<s$ and $z\in\mathcal D_{r',s'}$, then $\bar P(z;s-s',r-r')\subset\mathcal D_{r,s}$.
Source: Pos93 p. 213 ("analytic on complex neighbourhoods of a fixed domain $D\times\mathbb T^n$"); ours (max norms).

Proof. **Step S5.1: Openness and monotonicity.** $V_rD=\bigcup_{I'\in D}P(I',r)$ is a union of open polydiscs, and T_s is open; monotonicity is clear.

Step S5.2: Polydisc. Let $|\theta'-\theta|_\infty\le s-s'$ and $|I'-I|_\infty\le r-r'$. Then $|\operatorname{Im}\theta_j'|\le|\operatorname{Im}\theta_j|+s-s'<s$; and with $I_0\in D$, $|I-I_0|<r'$: $|I'-I_0|<r'+(r-r')=r$. □

Definition 5.2 (The class $\mathcal O(r,s)$ and the weighted norm)**.** $\mathcal O(r,s)$ is the set of $u:\mathcal D_{r,s}\to\mathbb C$ that are holomorphic (complex differentiable on the open set $\mathcal D_{r,s}\subset\mathbb C^{n+m}$) and 1-periodic in each θ_j. For $u\in\mathcal O(r,s)$ and $k\in\mathbb Z^n$, $I\in V_rD$, put

$$\hat u_k(I)=\int_{[0,1]^n}u(x,I)\,e^{-2\pi ik\cdot x}\,dx,\qquad |u|_{I,s}=\sum_{k\in\mathbb Z^n}|\hat u_k(I)|\,e^{2\pi|k|_1 s},\qquad |u|_{r,s}=\sup_{I\in V_rD}|u|_{I,s}\in[0,\infty].$$

(This is $|u|_{D,r,s}$; the integrand is continuous because $\mathbb R^n\subset T_s$.) For a vector- or matrix-valued $u=(u_a)_a$ (components in $\mathcal O(r,s)$) the coefficients are taken componentwise and $|\hat u_k|_\infty=\max_a|\hat u_{a,k}|$ replaces $|\hat u_k|$ in $|u|_{I,s}$. If $|u|_{r,s}<\infty$ we say u has *finite norm*.
Source: Pos93 p. 213 (definition of $|u|_{I,s}$, $|u|_{r,s}$), in the frozen convention of `docs/fourier-conventions.md` ("Bridge to strip sup norms": weight $e^{2\pi|k|_1 s}$).

Monotonicity: $|u|_{r',s'}\le|u|_{r,s}$ for $r'\le r$, $s'\le s$ (restriction to $\mathcal D_{r',s'}$; the coefficients at $I\in V_{r'}D$ are unchanged). For a vector, $\max_a|u_a|_{r,s}\le|u|_{r,s}\le\sum_a|u_a|_{r,s}$.

Lemma 5.3 (Coefficients, derivatives, series)**.** *Let $u\in\mathcal O(r,s)$.*

- *(a) Each $\hat u_k$ is holomorphic on V_rD.*
- *(b) $\partial_{\theta_j}u$, $\partial_{I_i}u\in\mathcal O(r,s)$, $\widehat{(\partial_{\theta_j}u)}_k=2\pi ik_j\hat u_k$, and $\widehat{(\partial_{I_i}u)}_k=\partial_{I_i}\hat u_k$.*
- *(c) (Series) Let $a_k:V_rD\to\mathbb C$ ($k\in\mathbb Z^n$) be holomorphic with $S:=\sup_{I\in V_rD}\sum_k|a_k(I)|e^{2\pi|k|_1 s}<\infty$. Then $v=\sum_k a_k\chi_k$ converges absolutely and uniformly on $\mathcal D_{r,s'}$ for each $s'<s$, $v\in\mathcal O(r,s)$, $\hat v_k=a_k$ and $|v|_{r,s}=S$.*
- *(d) (Representation) If $|u|_{r,s}<\infty$ then $u(\theta,I)=\sum_k\hat u_k(I)\chi_k(\theta)$ on $\mathcal D_{r,s}$, and $|\hat u_k(I)|e^{2\pi|k|_1 s}\le|u|_{r,s}$.*

Source: Standard; the one-variable ingredients are [K:`bp:p1:fourier-basic`], [K:`bp:p1:summation`].

Proof. **Step S5.3: (a).** Fix $I_0\in V_rD$ and $\eta>0$ with $\bar P(I_0,\eta)\subset V_rD$. The integrand $G(x,I)=u(x,I)e^{-2\pi ik\cdot x}$ is continuous on $[0,1]^n\times P(I_0,\eta)$, holomorphic in I, and D_IG is continuous there, because Du is holomorphic ([K:`bp:p1:holomorphic-regularity`]) hence continuous. (E1) gives holomorphy of $\hat u_k$ near I_0.

Step S5.4: (b). The derivatives are holomorphic ([K:`bp:p1:holomorphic-regularity`]) and periodic (differentiate the identity $u(\theta+e_j,I)=u(\theta,I)$). The formula for θ_j is [K:`bp:p1:fourier-basic`](a) at fixed I (the restriction of $u(\cdot,I)$ to $\mathbb R^n$ is continuous, periodic and C^1). The formula for I_i is the derivative formula of (E1) as in Step S5.3.

Step S5.5: (c). Each hypothesis $|a_k(I)|e^{2\pi|k|_1 s}\le S$ gives, on $\mathcal D_{r,s'}$, $|a_k(I)\chi_k(\theta)|\le|a_k(I)|e^{2\pi|k|_1 s'}\le Se^{-2\pi|k|_1(s-s')}$, which is summable ([K:`bp:p1:exp-sums`](b)). By the Weierstrass test the series converges uniformly on $\mathcal D_{r,s'}$, and each term is holomorphic, so v is holomorphic there ((E2)); as $s'<s$ is arbitrary, v is holomorphic on $\mathcal D_{r,s}$. It is periodic since each χ_k is. Integrating termwise ((E3)) against χ_{-k} over $[0,1]^n$ gives $\hat v_k=a_k$ (as in the coefficient step of [K:`bp:p1:summation`]); the norm identity is then the definition.

Step S5.6: (d). The bound is a single term of the sum $|u|_{I,s} \le |u|_{r,s}$. Fix I. The function $v = \sum_k \hat{u}_k(I)\chi_k$ is holomorphic and 1-periodic on T_s with coefficients $\hat{u}_k(I)$ ([K:bp:p1:summation], first part), and $u(\cdot, I) - v$ is holomorphic, 1-periodic, with all coefficients zero, so $u(\cdot, I) = v$ by [K:bp:p1:fourier-basic](c). □

5.3. **Lemma B.1: sup norm versus weighted norm.**

Lemma 5.4 (B.1). *Let $u \in \mathcal{O}(r, s)$ (scalar, vector or matrix valued;* $\sup|u|$ *is the sup of the max norm of the values).*

(a) $\sup_{\mathcal{D}_{r,s}} |u| \le |u|_{r,s}$.

(b) If $\sigma > 0$ and $|u| \le M$ on $\mathcal{D}_{r,s+\sigma}$ (so $u \in \mathcal{O}(r, s+\sigma)$), then $|u|_{r,s} \le \coth(\pi\sigma)^n M$.

In Pöschel's form: $|u|^{\sup}_{r,s} \le |u|_{r,s} \le \coth^n \sigma_P \, |u|^{\sup}_{r,s+2\sigma_P}$ with $\sigma_P = \pi\sigma$ (the coefficient radius grows by $2\sigma_P = 2\pi\sigma$); see Remark 5.10.

Source: Pos93 p. 213, Lemma B.1; frozen-convention form: [K:bp:p1:summation], [K:bp:p1:decay], [K:bp:p1:exp-sums]*(b); this is Lemma 2.9 with the I-dependence made explicit.*

Proof. **Step S5.7: (a).** If $|u|_{r,s} = \infty$ there is nothing to prove. Otherwise Lemma 5.3(d) gives, for $(\theta, I) \in \mathcal{D}_{r,s}$, $|u(\theta, I)| \le \sum_k |\hat{u}_k(I)||\chi_k(\theta)| \le \sum_k |\hat{u}_k(I)| e^{2\pi|k|_1 s}$, using $|\chi_k(\theta)| \le e^{2\pi|k|_1 |\mathrm{Im}\,\theta|_\infty}$ ([K:bp:p0:domains]). For vector values apply this to each component and note $|\hat{u}_{a,k}| \le |\hat{u}_k|_\infty$.

Step S5.8: (b). Fix $I \in V_r D$. The function $u(\cdot, I)$ is holomorphic, periodic and bounded by M on $T_{s+\sigma}$, so [K:bp:p1:decay] gives $|\hat{u}_k(I)|_\infty \le e^{-2\pi|k|_1(s+\sigma)} M$ (limit $\rho' \uparrow s+\sigma$), and

$$|u|_{I,s} \le M \sum_k e^{-2\pi|k|_1 \sigma} = \coth(\pi\sigma)^n M \qquad ([\text{K:bp:p1:exp-sums}](b),\ x = 2\pi\sigma).$$

Take the supremum over I. □

5.4. **Lemma B.2: algebra property.**

Lemma 5.5 (B.2). *Let $u, v \in \mathcal{O}(r, s)$ be scalar with finite norms. Then $uv \in \mathcal{O}(r, s)$, $\widehat{(uv)}_k(I) = \sum_l \hat{u}_{k-l}(I)\hat{v}_l(I)$ (absolutely convergent), and $|uv|_{I,s} \le |u|_{I,s}\,|v|_{I,s}$ for every $I \in V_r D$; hence $|uv|_{r,s} \le |u|_{r,s}|v|_{r,s}$, and $|u^j|_{r,s} \le |u|^j_{r,s}$.*

Source: Pos93 p. 213, Lemma B.2. (The coefficient-level inequality is proved in Lean: `docs/fourier-conventions.md`, "Verified coefficient multiplication", `KAM/Fourier/Convolution.lean`; its identification with the product of functions is the remaining bridge.)

Proof. **Step S5.9: Product is in the class.** uv is holomorphic and periodic.

Step S5.10: Coefficients. Fix I. By Lemma 5.3(d), at real x, $u(x, I) = \sum_k \hat{u}_k \chi_k(x)$ and $v(x, I) = \sum_l \hat{v}_l \chi_l(x)$, both absolutely convergent ((E3)). The Cauchy product gives $u(x, I)v(x, I) = \sum_k c_k \chi_k(x)$ with $c_k = \sum_l \hat{u}_{k-l}\hat{v}_l$, and $\sum_k |c_k| \le |u|_{I,0}|v|_{I,0} < \infty$, so the convergence is uniform in x and termwise integration gives $\widehat{(uv)}_k(I) = c_k$.

Step S5.11: Norm. Since $|k|_1 \le |k-l|_1 + |l|_1$, $|uv|_{I,s} = \sum_k |c_k| e^{2\pi|k|_1 s} \le \sum_k \sum_l |\hat{u}_{k-l}| e^{2\pi|k-l|_1 s} |\hat{v}_l| e^{2\pi|l|_1 s} = |u|_{I,s}|v|_{I,s}$ (Tonelli, with the bijection $(k, l) \mapsto (k-l, l)$). Take the supremum over I; induct for u^j. □

5.5. **Lemma B.3: Cauchy estimates in the weighted norm.**

Lemma 5.6 (B.3). *Let $u \in \mathcal{O}(r_0, s_0)$ have finite norm, $0 < r < r_0$, $0 < s < s_0$. Then*

(a) $\displaystyle\sum_{j=1}^{n} |\partial_{\theta_j} u|_{r_0,s} \le \frac{1}{e(s_0 - s)} |u|_{r_0,s_0}$;

(b) $\displaystyle\max_{1 \le i \le m} |\partial_{I_i} u|_{r,s_0} \le \frac{1}{r_0 - r} |u|_{r_0,s_0}$.

In particular (taking $r_0 = r+\rho$, $s_0 = s+\sigma$): $\sum_j |\partial_{\theta_j} u|_{r,s-\sigma} \le (e\sigma)^{-1}|u|_{r,s}$ and $\max_i |\partial_{I_i} u|_{r-\rho,s} \le \rho^{-1}|u|_{r,s}$ for $0<\sigma<s$, $0<\rho<r$. The same holds for vector- and matrix-valued u, componentwise.
Source: Pos93 pp. 213–214, Lemma B.3 (with s geometric; Pöschel's $1/(e\sigma)$ is the same number, Remark 5.10).

Proof. **Step S5.12: (a): coefficients.** Fix $I \in V_{r_0}D$. By Lemma 5.3(b), $\sum_j |\partial_{\theta_j} u|_{I,s} = \sum_k \sum_j 2\pi|k_j||\hat{u}_k| e^{2\pi|k|_1 s} = \sum_k 2\pi|k|_1 e^{-2\pi|k|_1(s_0-s)}|\hat{u}_k| e^{2\pi|k|_1 s_0}$. By [K:`bp:p1:exp-sums`](a) with $\tau = 1$, $a = 2\pi(s_0-s)$: $2\pi|k|_1 e^{-2\pi|k|_1(s_0-s)} \le 2\pi/(e\cdot 2\pi(s_0-s))$. Hence the sum is $\le (e(s_0-s))^{-1}|u|_{I,s_0}$; take the supremum over I.

Step S5.13: (b): Cauchy integral. Fix $I \in V_r D$, i, and put $\rho' = r_0 - r$. For $|\zeta| \le \rho'$ the point $I + \zeta e_i$ lies in $V_{r_0}D$: if $I_0 \in D$ with $|I - I_0| < r$ then $|I + \zeta e_i - I_0| < r + \rho' = r_0$. By Lemma 5.3(a),(b) and [K:`bp:p1:ext-cauchy`], $\partial_{I_i}\hat{u}_k(I) = \frac{1}{2\pi i}\oint_{|\zeta|=\rho'} \hat{u}_k(I+\zeta e_i)\zeta^{-2}\,d\zeta$, so $|\partial_{I_i}\hat{u}_k(I)| \le \frac{1}{2\pi}\oint |\hat{u}_k(I+\zeta e_i)|\rho'^{-2}|d\zeta|$. Multiply by $e^{2\pi|k|_1 s_0}$, sum over k and use Tonelli ((E3)): $|\partial_{I_i} u|_{I,s_0} \le \frac{1}{2\pi}\oint \rho'^{-2}|u|_{I+\zeta e_i,s_0}|d\zeta| \le \rho'^{-1}|u|_{r_0,s_0}$. The vector case is componentwise, the matrix case likewise. □

5.6. **Lemma B.4: the Poisson bracket.** For $m = n$ and u, v holomorphic on an open subset of $\mathbb{C}^n \times \mathbb{C}^n$, $\{u,v\} = \sum_{i=1}^n \big(\partial_{\theta_i} u\,\partial_{I_i} v - \partial_{I_i} u\,\partial_{\theta_i} v\big)$ (Definition 6.2; $\{u,v\} = Du \cdot X_v$ for the frozen field $X_v = (\partial_I v, -\partial_\theta v)$).

Lemma 5.7 (B.4)**.** *Let $m = n$, $u \in \mathcal{O}(r_0,s_0)$, $v \in \mathcal{O}(r,s)$ with finite norms, $0<\rho<r$, $0<\sigma<s$, $r-\rho<r_0$, $s-\sigma<s_0$. Then $\{u,v\} \in \mathcal{O}(r-\rho, s-\sigma)$ and*

$$|\{u,v\}|_{r-\rho,s-\sigma} \le \frac{1}{e}\Big(\frac{1}{(r_0-r+\rho)\,\sigma} + \frac{1}{(s_0-s+\sigma)\,\rho}\Big)|u|_{r_0,s_0}|v|_{r,s}.$$

Source: Pos93 p. 214, Lemma B.4 (same formula, with Pöschel's $(r_0,s_0,r,s,\rho,\sigma)$ read as our geometric quantities).

Proof. Write $\{u,v\} = \sum_i \partial_{\theta_i} u\,\partial_{I_i} v - \sum_i \partial_{I_i} u\,\partial_{\theta_i} v$, all terms restricted to $\mathcal{D}_{r-\rho,s-\sigma} \subset \mathcal{D}_{r_0,s_0} \cap \mathcal{D}_{r,s}$ (monotonicity).

Step S5.14: Four derivative bounds. By Lemma 5.6 and monotonicity of the norm in both widths: $\sum_i |\partial_{\theta_i} u|_{r-\rho,s-\sigma} \le \frac{|u|_{r_0,s_0}}{e(s_0-s+\sigma)}$ ((a) with $s \to s-\sigma$, $r_0 \to r_0$); $\max_i |\partial_{I_i} u|_{r-\rho,s-\sigma} \le \frac{|u|_{r_0,s_0}}{r_0-r+\rho}$ ((b) with $r \to r-\rho$); $\sum_i |\partial_{\theta_i} v|_{r-\rho,s-\sigma} \le \frac{|v|_{r,s}}{e\sigma}$ and $\max_i |\partial_{I_i} v|_{r-\rho,s-\sigma} \le \frac{|v|_{r,s}}{\rho}$.

Step S5.15: Products. By Lemma 5.5 and the triangle inequality for $|\cdot|_{I,s-\sigma}$ (coefficients of a sum are sums), $|\{u,v\}|_{r-\rho,s-\sigma} \le \sum_i |\partial_{\theta_i} u|_,|\partial_{I_i} v|_, + \sum_i |\partial_{I_i} u|_,|\partial_{\theta_i} v|_, \le \big(\sum_i |\partial_{\theta_i} u|_,\big)\max_i |\partial_{I_i} v|_, + \max_i |\partial_{I_i} u|_,\big(\sum_i |\partial_{\theta_i} v|_,\big)$, and inserting the four bounds gives the claim. □

5.7. **Lemma C.1: the general Cauchy inequality.**

Lemma 5.8 (C.1)**.** *Let A, B be finite-dimensional complex normed spaces, $v \in A$, $R>0$, and $G : P_A(v,R) = \{a : |a-v|_A < R\} \to B$ holomorphic with $|G|_B \le M$ on this ball. Then the operator norm satisfies $|DG(v)|_{B,A} \le M/R$.*
Source: Pos93 p. 215, Lemma C.1 ("analytic maps of Banach spaces"; only the finite-dimensional case is needed).

Proof. **Step S5.16: Reduction to one variable.** Let $u \in A$, $u \ne 0$. $f(\zeta) = G(v+\zeta u)$ is holomorphic on $|\zeta| < R/|u|_A$ with $|f| \le M$ and $f'(0) = DG(v)u$.

Step S5.17: Cauchy. For every $\varrho < R/|u|_A$, [K:`bp:p1:ext-cauchy`] gives $|f'(0)| \le M/\varrho$; let $\varrho \uparrow R/|u|_A$: $|DG(v)u|_B \le M|u|_A/R$. □

This also gives (E2)'s derivative statement: apply it to $K_j - K_l$ on the ball of radius η.

5.8. **Projections and truncation.**

Lemma 5.9 (P_S, T_K, averages)**.** *Let $u \in \mathcal{O}(r,s)$ have finite norm and $S \subset \mathbb{Z}^n$. Put $P_S u = \sum_{k\in S} \hat{u}_k \chi_k$. Then $P_S u \in \mathcal{O}(r,s)$, $\widehat{(P_S u)}_k = \hat{u}_k$ for $k \in S$ and 0 otherwise, and $|P_S u|_{r,s} \le |u|_{r,s}$. In particular, for $K \ge 0$ the truncation $T_K u = P_{\{|k|_1 \le K\}} u$ (a trigonometric polynomial in θ) and the average $\langle u\rangle(I) = \hat{u}_0(I) = P_{\{0\}} u$ (independent of θ) satisfy $|T_K u|_{r,s} \le |u|_{r,s}$, $|\langle u\rangle|_{r,s} \le |u|_{r,s}$, and*

$$u - T_K u = P_{\{|k|_1 > K\}} u, \qquad |u - T_K u|_{r,s-\sigma} \le e^{-2\pi K\sigma} |u|_{r,s} \qquad (0 < \sigma < s).$$

If $\Lambda \subset \mathbb{Z}^n$ is any set (e.g. a lattice, Pos93 p. 192) and u is said to be in Λ-normal form when $u = P_\Lambda u$, then $P_\Lambda u$, $u - P_\Lambda u$ are in the class with norms $\le |u|_{r,s}$. For vector-valued u everything is componentwise.

Source: Pos93 p. 192 (definition of T_K, P_Λ) and p. 212 ("$|f - T_K f|_{r,s-\sigma} \le e^{-K\sigma}|f|_{r,s}$"), geometric form.

Proof. **Step S5.18: Class and coefficients.** The coefficient family $a_k = \hat{u}_k \mathbf{1}_S(k)$ consists of holomorphic functions (Lemma 5.3(a)) with $\sup_I \sum_k |a_k| e^{2\pi|k|_1 s} \le |u|_{r,s} < \infty$; Lemma 5.3(c) gives all claims including $|P_S u|_{r,s} \le |u|_{r,s}$.

Step S5.19: Tail. For $|k|_1 > K$: $e^{2\pi|k|_1(s-\sigma)} = e^{-2\pi|k|_1\sigma} e^{2\pi|k|_1 s} \le e^{-2\pi K\sigma} e^{2\pi|k|_1 s}$. Sum over k with $|k|_1 > K$ and take the supremum over I. Also $u = \sum_k \hat{u}_k \chi_k$ (Lemma 5.3(d)) gives $u - T_K u = P_{\{|k|_1 > K\}} u$. □

Remark 5.10 (Dictionary with Pöschel's normalisation and with Part 1). Pöschel works with 2π-periodic angles φ, characters $e^{ik\cdot\varphi}$, weight $e^{|k|_1 s_P}$ and losses (ρ_P, σ_P). Under the symplectic scaling $\varphi = 2\pi\theta$, $I_P = I/2\pi$ (so $d\varphi \wedge dI_P = d\theta \wedge dI$; time and the values of Hamiltonians are unchanged) his objects become ours with $s_P = 2\pi s$, $\sigma_P = 2\pi\sigma$, $r_P = r/2\pi$, $\rho_P = \rho/2\pi$. Then $e^{|k|_1 s_P} = e^{2\pi|k|_1 s}$ and $\rho_P\sigma_P = \rho\sigma$; the quantities in which Lemmas B.3–B.5 and the Iterative Lemma are written (for instance $|\phi|/(\rho\sigma)$ and the norm $|W\cdot|$) are therefore the same numbers. The conversion of Part 1, $\tilde{H} = 2\pi H(\varphi/2\pi, I)$ with I unchanged (Lemma 2.8), gives the same smallness ratio: $\tilde{\phi}/(\rho\sigma_P) = 2\pi\phi/(\rho \cdot 2\pi\sigma) = \phi/(\rho\sigma)$. So a condition "$|\phi| < \rho\sigma/2$" means the same in all three conventions, and the 2π-conversion does not affect the constants of the Iterative Lemma beyond the scalings of Lemma 2.8(b) (that bookkeeping is Part 6's). Where `docs/fourier-conventions.md` writes the derivative loss as $2\pi/(e\delta)$ for coefficient radius s, Lemma 5.6(a) is the same statement for the geometric width: a coefficient-radius loss $\delta = 2\pi\sigma$ gives $2\pi/(e \cdot 2\pi\sigma) = 1/(e\sigma)$.

6. Canonical transformations: time-one maps and the Lie transform (shared by tracks N, P)

This part is the canonical-transformation toolkit of Pöschel 1993, Appendix A (pp. 211–213: Taylor's formula, the flow X_ϕ^t of a small Hamiltonian field, its displacement and derivative bounds) and Lemma B.5 (pp. 214–215), in the frozen conventions of Part 5 (period 1, geometric widths, max norms). Track P cites it as [N:`bpN:p5:...`]. Generating functions ϕ live on general domains $\mathcal{D}_{r,s} = T_s \times V_r D$ with $D \subset \mathbb{C}^n$ arbitrary (Definition 5.1); $m = n$ because a Hamiltonian structure is needed. The ODE existence, uniqueness and continuation used are those of Part 4 (general layer); the weighted-norm estimates are those of Part 5.

Remark 6.1 (Decision for Lemma B.5: the Lie series, not the sup-norm fallback). **Route.** `PLAN.md` left open whether $|u \circ X_\phi^1| \le (1 - 2|\phi|/(\rho\sigma))^{-1}|u|$ is proved by Pöschel's convergent series $u \circ X_\phi^1 = \sum_h \frac{1}{h!} \operatorname{ad}_\phi^h u$ or by a sup-norm estimate of the displaced function and a conversion back with Lemma 5.4(b). *We take the series*, for three reasons. (1) The Normal Form Lemma iterates $N \approx Ks/(12\ln 2)$ times and every step feeds the *weighted* norm of f_+ into the next step. The fallback re-enters the weighted norm through $\coth(\pi\sigma)^n$ in every step; this factor exceeds 1 by a fixed amount, so the contraction $\varepsilon_{i+1} \le \frac{1}{4}\varepsilon_i$ of the Iterative Lemma would be destroyed and not

merely perturbed in constants. (2) The fallback needs $u \circ X^1_\phi$ to be holomorphic and periodic *before* one can speak of its Fourier coefficients, that is, holomorphic dependence of the flow on the initial point (Lemma 6.4(b)), a statement not in Mathlib. The series route gets holomorphy of $u \circ X^1_\phi$ for free (it is a norm-convergent series of holomorphic functions); the dependence on initial data is then needed only for the claims on the map $\Phi = X^1_\phi$ itself (Proposition 6.13). (3) The ingredients of the series route are already labelled results: the bracket estimate (Lemma 5.7), a one-line induction (Lemma 6.9), a Taylor remainder along a solution of $\dot z = X_\phi(z)$ (Part 4 and Lemma 6.8) and a geometric series. **What is lost.** The equality $u \circ X^1_\phi = \sum_h \frac{1}{h!} \operatorname{ad}^h_\phi u$ is proved here for $\eta := 2|\phi|/(\rho\sigma) \le \frac12$ instead of Pöschel's $\eta < 1$ (Theorem 6.10); the bound itself is $(1-\eta)^{-1}$ as in the source. For $\eta < 1$ the equality also holds, by real-analyticity of $t \mapsto u(X^t_\phi z)$ and the identity theorem, but real-analytic dependence on time is another external fact; since the frozen headline only asserts the existence of constants, Part 6 simply uses $|\phi| \le \rho\sigma/4$ in place of $|\phi| < \rho\sigma/2$.

6.1. **Setting.**

Definition 6.2 (Hamiltonian field, bracket, scaled norm)**.** Points of $\mathbb{C}^n \times \mathbb{C}^n$ are $z = (\theta, I)$ (angles first, as in the frozen field); $J = \begin{pmatrix} 0 & \mathbf{1} \\ -\mathbf{1} & 0 \end{pmatrix}$, so $J^2 = -\mathbf{1}$, $J^\top = -J$. For u holomorphic on an open set let $\nabla u = (\partial_\theta u, \partial_I u)^\top$,

$$X_u = J\nabla u = (\partial_I u, -\partial_\theta u), \qquad \{u, v\} = \sum_{i=1}^{n} \big(\partial_{\theta_i} u\, \partial_{I_i} v - \partial_{I_i} u\, \partial_{\theta_i} v\big) = Du \cdot X_v = \nabla u^\top J \nabla v = -\{v, u\}.$$

X_u is the frozen Hamiltonian field (`KAM.hamiltonianVectorField`, Definition 1.3; for a real-valued function on real points see [K:`bp:p1:hamiltonian-real`]); the bracket is normalised so that $\frac{d}{dt} u(X^t_\phi(z)) = \{u, \phi\}(X^t_\phi(z))$ (Lemma 6.7). For $\rho, \sigma > 0$ the *scaled norm* on $\mathbb{C}^n \times \mathbb{C}^n$ is $\nu_{\rho,\sigma}(a, b) = \max(|a|_\infty/\sigma, |b|_\infty/\rho)$ (Pöschel's $|W\,\cdot|$, $W = \operatorname{diag}(\sigma^{-1}, \rho^{-1})$ in angle-first order); it is a complex norm, and its closed ball of radius c about $z = (\theta, I)$ is $\bar P(z; c\sigma, c\rho)$. Iterated brackets: $\operatorname{ad}^0_\phi u = u$, $\operatorname{ad}^j_\phi u = \{\operatorname{ad}^{j-1}_\phi u, \phi\}$.

Source: Pos93 pp. 211–212 (flow X^t_ϕ, W), p. 214 (ad).

Source gap: We did not locate the printed definition of the bracket's sign in pp. 211–215; only the two-term structure (Lemma B.4) and $\{\phi, h\} = \omega \cdot \partial_\theta \phi$ for $h = h(I)$ (p. 211, homological equation, divisors $ik \cdot \omega$) are used. With the convention above $\{\phi, h\} = \partial_\theta \phi \cdot \partial_I h$, which is that equation.

The bracket of holomorphic periodic functions is holomorphic and periodic ([K:`bp:p1:holomorphic-regularity`]); $\operatorname{ad}^j_\phi u$ is holomorphic on the common domain of u and ϕ; $\{u, v\} = 0$ if u, v depend only on I.

Lemma 6.3 (Size of the field)**.** *Let $\phi \in \mathcal{O}(r, s)$ have finite norm, $0 < \rho < r$, $0 < \sigma < s$. On $\mathcal{D}_{r-\rho, s-\sigma}$: $|\partial_{I_i}\phi| \le |\phi|_{r,s}/\rho$, $|\partial_{\theta_i}\phi| \le |\phi|_{r,s}/(e\sigma)$, and*

$$\nu_{\rho,\sigma}(X_\phi) = \max\Big(\frac{\max_i |\partial_{I_i}\phi|}{\sigma}, \frac{\max_i |\partial_{\theta_i}\phi|}{\rho}\Big) \le \frac{|\phi|_{r,s}}{\rho\sigma}.$$

Source: Pos93 p. 212 ("$|\phi_I|_\infty \le q\delta/(\alpha\rho)$, $|WX_\phi| \le q\delta/(\alpha\rho\sigma)$").

Proof. **Step S6.1: Derivatives.** Lemma 5.6 (with $r_0 = r$, $s_0 = s$ and the target widths $r - \rho$, $s - \sigma$) gives $\max_i |\partial_{I_i}\phi|_{r-\rho,s} \le \rho^{-1}|\phi|_{r,s}$ and $\sum_j |\partial_{\theta_j}\phi|_{r,s-\sigma} \le (e\sigma)^{-1}|\phi|_{r,s}$; each single term is at most the sum. Restrict to $\mathcal{D}_{r-\rho,s-\sigma}$ (monotonicity), and apply Lemma 5.4(a) to each derivative.

Step S6.2: The field. Insert into the definition of $\nu_{\rho,\sigma}$: $\frac{1}{\sigma} \cdot \frac{|\phi|}{\rho} = \frac{|\phi|}{\rho\sigma}$ and $\frac{1}{\rho} \cdot \frac{|\phi|}{e\sigma} \le \frac{|\phi|}{\rho\sigma}$. □

6.2. **The flow of a small holomorphic vector field.**

Lemma 6.4 (Flow of a small holomorphic field)**.** *Let ν be a complex norm on $\mathbb{C}^N$, $\Omega \subset \mathbb{C}^N$ open, $F : \Omega \to \mathbb{C}^N$ holomorphic with $\nu(F) \le \varepsilon$ on Ω and $0 \le \varepsilon \le \frac{1}{2}$. For $c > 0$ let $\Omega^c = \{z : \bar{B}_\nu(z, c) \subset \Omega\}$ (an open set).*

(a) (Existence, uniqueness, displacement) For $z \in \Omega^{1/2}$ there is $\gamma_z : \mathbb{R} \to \mathbb{C}^N$ with $\mathtt{IsSolIn}\, F\, \Omega\, [-1,1]\, \gamma_z$ and $\gamma_z(0) = z$; any solution in Ω through z on an interval $J \ni 0$ equals γ_z on $J \cap [-1,1]$. Put $X^t(z) = \gamma_z(t)$ ($|t| \le 1$). Then $\nu(X^t(z) - z) \le |t|\,\varepsilon$.

(b) (Regularity) $(t,z) \mapsto X^t(z)$ is continuous on $[-1,1]\times\Omega^{1/2}$; for each t the map $X^t : \Omega^{1/2} \to \Omega$ is holomorphic; $(t,z) \mapsto D_z X^t(z)$ is continuous, and $Y(t) = D_z X^t(z)$ satisfies $Y(t) = \mathbf{1} + \int_0^t DF(X^\tau z) Y(\tau)\, d\tau$.

(c) (Inverse, derivative) For $z \in \Omega^1$: $X^1(z) \in \Omega^{1/2}$ and $X^{-1}(X^1 z) = z$; and the ν-operator norm satisfies $|DX^t(z) - \mathbf{1}|_\nu \le 2\varepsilon$ for $|t| \le 1$. In the same way $X^1(X^{-1}z) = z$.

(d) (Translations) If $\tau \in \mathbb{C}^N$, $\Omega + \tau = \Omega$ and $F(w+\tau) = F(w)$ on Ω, then $\Omega^{1/2} + \tau = \Omega^{1/2}$ and $X^t(z+\tau) = X^t(z) + \tau$.

(e) (Reality) Let $N = 2n$ and suppose $F(x) \in \mathbb{R}^N$ for $x \in \Omega \cap \mathbb{R}^N$. Then $X^t(x) \in \mathbb{R}^N$ for $x \in \Omega^{1/2} \cap \mathbb{R}^N$.

Source: Pos93 p. 212 ("X^t_ϕ maps $V_{r-\frac{3}{2}\rho, s-\frac{3}{2}\sigma}$ into $V_{r-\rho,s-\sigma}$ for $0 \le t \le 1$ and satisfies $|W(X^t_\phi - \mathrm{id})| \le \ldots$"; the derivative estimate by Lemma C.1). The general-ν formulation and the holomorphy in (b) are ours (Pöschel asserts "real analytic").

Proof. Regard $\mathbb{C}^N$ as a real normed space of dimension $2N$ with norm ν, extend F by 0 outside Ω (Remark 4.1); F is C^1 on Ω ([K:`bp:p1:holomorphic-regularity`], restriction of scalars).

Step S6.3: (a) The maximal solution stays in a compact set. Let $z \in \Omega^{1/2}$, (T_-, T_+) and γ the maximal interval and solution (Theorem 4.13). For $t \in [0, T_+)$, $\nu(\gamma'(t)) \le \varepsilon$, so $\nu(\gamma(t) - z) \le \varepsilon t$ ([K:`bp:p1:ext-mvt`] in one variable). If $T_+ \le 1$, then $\gamma(t) \in K = \bar{B}_\nu(z, \varepsilon) \subset \bar{B}_\nu(z, \frac{1}{2}) \subset \Omega$ for $t \in [0, T_+)$, with K compact, so $T_+ = \infty$ by Corollary 4.15, a contradiction. Hence $T_+ > 1$; symmetrically (time reversal, Lemma 4.3(b)) $T_- > -1$. Uniqueness on an interval $J \ni 0$ is Theorem 4.11; the displacement bound was shown.

Step S6.4: (b) Picard iterates. Fix $z_0 \in \Omega^{1/2}$. As $\bar{B}_\nu(z_0, \frac{1}{2})$ is compact in the open Ω, there is $\eta > 0$ with $\bar{B}_\nu(z_0, \frac{1}{2} + \eta) \subset \Omega$. Let $\Lambda = \sup\{|DF(w)|_\nu : w \in \bar{B}_\nu(z_0, \frac{1}{2} + \eta)\} < \infty$; F is Λ-Lipschitz on this convex ball ([K:`bp:p1:ext-mvt`]). For $(t,z) \in [-1,1] \times B_\nu(z_0, \eta)$ put $y_0(t,z) = z$, $y_{k+1}(t,z) = z + \int_0^t F(y_k(\tau, z))\, d\tau$. By induction on k: $\nu(y_k(t,z) - z) \le |t|\varepsilon$ (so $y_k(t,z) \in \bar{B}_\nu(z,\varepsilon) \subset \bar{B}_\nu(z_0, \frac{1}{2} + \eta) \subset \Omega$, and the integral is defined); y_k is continuous; $y_k(t,\cdot)$ is holomorphic; $D_z y_k$ is continuous in (t,z). The last two follow from (E1) applied to $t\int_0^1 F(y_k(t\tau', z))\, d\tau'$, whose z-derivative is $t\int_0^1 DF(y_k) D_z y_k\, d\tau'$.

Step S6.5: (b) Convergence and identification. $\nu(y_{k+1} - y_k)(t,z) \le \varepsilon \Lambda^k |t|^{k+1}/(k+1)!$ by induction, so $y_k \to y_\infty$ uniformly on $[-1,1] \times B_\nu(z_0, \eta)$. The limit is continuous and satisfies $y_\infty(t,z) = z + \int_0^t F(y_\infty)\, d\tau$ (F is Λ-Lipschitz), so $y_\infty(\cdot, z)$ is C^1 with derivative $F(y_\infty)$, i.e. a solution in Ω on $[-1,1]$ through z, and $y_\infty(t,z) = X^t(z)$ by (a). Hence X^t is continuous in (t,z) and, by (E2), holomorphic on $B_\nu(z_0,\eta)$ with $D_z y_k \to D_z X^t$ uniformly on $[-1,1] \times B_\nu(z_0, \eta/2)$, so $D_z X^t(z)$ is jointly continuous. As z_0 was arbitrary, (b) holds except for the integral equation. Differentiating $X^t(w) = w + \int_0^t F(X^\tau w)\, d\tau$ in w under the integral ((E1); the integrand's w-derivative $DF(X^\tau w) D_w X^\tau(w)$ is jointly continuous) gives the equation for Y.

Step S6.6: (c). Let $z \in \Omega^1$, $w = X^1(z)$. Then $\nu(w - z) \le \varepsilon \le \frac{1}{2}$, so $\bar{B}_\nu(w, \frac{1}{2}) \subset \bar{B}_\nu(z, 1) \subset \Omega$, i.e. $w \in \Omega^{1/2}$. The map $t \mapsto \gamma_z(1+t)$ is a solution on $[-1,0]$ through w, so it equals γ_w there (Theorem 4.11) and $X^{-1}(w) = \gamma_z(0) = z$. The statement for $X^1 \circ X^{-1}$ is this one for $-F$

(Lemma 4.3(b)). For the derivative: $B_\nu(z,\frac{1}{2}) \subset \Omega^{1/2}$ (as above), $G = X^t - \mathrm{id}$ is holomorphic there by (b) with $\nu(G) \le \varepsilon$ by (a); Lemma 5.8 with $R = \frac{1}{2}$ gives $|DG(z)|_\nu \le 2\varepsilon$.

Step S6.7: (d),(e). (d): $\Omega^{1/2} + \tau = \Omega^{1/2}$ since the ν-balls translate; $t \mapsto \gamma_z(t) + \tau$ is a solution through $z+\tau$ (Lemma 4.3(c)), so equals $\gamma_{z+\tau}$ by uniqueness. (e): apply (a) to $E = \mathbb{R}^N$, $W = \Omega \cap \mathbb{R}^N$, the real field $F|_W$ (it is C^1 and real by hypothesis, [K:`bp:p1:real-restriction`]) and the compact set $\bar{B}_\nu(x,\varepsilon) \cap \mathbb{R}^N \subset W$: this gives a real solution on $[-1,1]$ through x, which is also a solution in Ω, hence equals γ_x. □

Lemma 6.5 (Symplectic flow)**.** *In Lemma 6.4 let $N = 2n$, $\Omega \subset \mathbb{C}^n \times \mathbb{C}^n$, and $F = X_\phi$ with ϕ holomorphic on Ω. Then for $z \in \Omega^{1/2}$, $|t| \le 1$, $Y = DX^t(z)$ satisfies*

$$Y^\top J Y = J, \qquad Y J Y^\top = J, \qquad \det Y = \pm 1$$

($Y^\top$ the plain transpose of the complex matrix).
Source: Standard (Hamiltonian flows are symplectic; used implicitly by Pos93 pp. 211–212, "canonical transformation Φ").

Proof. **Step S6.8: Variational equation.** By Lemma 6.4(b), $Y \in C^1$ with $Y' = DF(X^t z)Y$, $Y(0) = \mathbf{1}$.

Step S6.9: The field. $F = J\nabla\phi$, so $DF = JS$ with $S = D\nabla\phi$ the Hessian, symmetric ([K:`bp:p1:ext-schwarz`]). Then $DF^\top J = S^\top J^\top J = S$ and $J\,DF = J^2 S = -S$, so $DF^\top J + J\,DF = 0$.

Step S6.10: Conservation. $\frac{d}{dt}(Y^\top J Y) = Y^\top (DF^\top J + J\,DF) Y = 0$, so $Y^\top J Y = J$ for all t. Taking determinants, $\det(Y)^2 = 1$; Y is invertible and inverting $Y^\top J Y = J$ gives $Y^{-1} J^{-1} Y^{-\top} = J^{-1}$, that is, $J = Y J Y^\top$ ($J^{-1} = -J$). □

Corollary 6.6 (Canonical maps: bracket invariance and conjugation)**.** *In the setting of Lemma 6.5, let $z \in \Omega^{1/2}$, $\Phi = X^t$ ($|t| \le 1$), and u, v, H holomorphic on a neighbourhood of $\Phi(z)$. Then $u \circ \Phi$ is holomorphic near z and*

- *(a) $\{u \circ \Phi, v \circ \Phi\}(z) = \{u, v\}(\Phi(z))$;*
- *(b) $D\Phi(z)\, X_{H\circ\Phi}(z) = X_H(\Phi(z))$;*
- *(c) if $y : J \to \Omega^{1/2}$ satisfies $\dot y = X_{H\circ\Phi}(y)$ with H holomorphic on a neighbourhood of $\Phi(y(J))$, then $x = \Phi \circ y$ satisfies $\dot x = X_H(x)$.*

Source: Standard.

Proof. **Step S6.11: Chain rule.** Φ is holomorphic (Lemma 6.4(b)) and $\nabla(u \circ \Phi) = D\Phi^\top (\nabla u) \circ \Phi$.

Step S6.12: (a). $\{u \circ \Phi, v \circ \Phi\} = \nabla u^\top Y J Y^\top \nabla v = \nabla u^\top J \nabla v$ at $\Phi(z)$ by Lemma 6.5.

Step S6.13: (b),(c). $D\Phi\, X_{H\circ\Phi} = Y\, J\, Y^\top (\nabla H) \circ \Phi = J(\nabla H) \circ \Phi = X_H \circ \Phi$; (c) is the chain rule $\dot x = D\Phi(y)\dot y = D\Phi(y) X_{H\circ\Phi}(y)$. □

For real data: if ϕ is real on real points then so is $F = X_\phi$ (Step S6.7), Φ maps real points to real points, its derivative at a real point is a real matrix ([K:`bp:p1:real-restriction`]), and (a)–(c) hold for real u, v, H with the real restriction of X ([K:`bp:p1:hamiltonian-real`]).

6.3. Taylor's formula along the flow.

Lemma 6.7 (Derivative along the flow)**.** *Let $\Omega \subset \mathbb{C}^n \times \mathbb{C}^n$, ϕ, $F = X_\phi$, $\varepsilon \le \frac{1}{2}$ as in Lemma 6.5; let $\Omega_1 \subset \Omega$ be open and u holomorphic on Ω_1, and $z \in \Omega^{1/2}$ with $X^t(z) \in \Omega_1$ for $|t| \le 1$. Then $g(t) = u(X^t(z))$ is C^∞ on $[-1,1]$ and $g^{(j)}(t) = (\mathrm{ad}_\phi^j u)(X^t(z))$ for all $j \ge 0$ (each $\mathrm{ad}_\phi^j u$ is holomorphic on Ω_1).*
Source: Pos93 p. 211 ("Taylor's formula" for $h \circ X_\phi^t$); standard.

Proof. **Step S6.14: One derivative.** For v holomorphic on Ω_1, v is real-differentiable with derivative Dv, and $\gamma = X^{\cdot}(z)$ has derivative $X_\phi(\gamma)$, so $\frac{d}{dt}v(\gamma(t)) = Dv(\gamma)\cdot X_\phi(\gamma) = \{v,\phi\}(\gamma(t))$ (Definition 6.2), which is continuous in t.

Step S6.15: Induction. Apply the previous step to $v = \mathrm{ad}_\phi^j u$, holomorphic on Ω_1 (derivatives of holomorphic functions, [K:bp:p1:holomorphic-regularity]). □

Lemma 6.8 (Taylor with integral remainder along the flow)**.** *In the setting of Lemma 6.7, for every $N \ge 1$,*

$$u(X^1 z) = \sum_{j=0}^{N-1} \frac{1}{j!}(\mathrm{ad}_\phi^j u)(z) + R_N(z),$$

$$R_N(z) = \int_0^1 \frac{(1-t)^{N-1}}{(N-1)!}(\mathrm{ad}_\phi^N u)(X^t z)\,dt, \qquad |R_N(z)| \le \frac{1}{N!}\max_{0\le t\le 1}|(\mathrm{ad}_\phi^N u)(X^t z)|.$$

In particular ($N = 1$ for f_0, $N = 2$ for h), if h, f_0 are holomorphic on Ω_1 and the trajectory of z stays in Ω_1,

$$(h+f_0)(X^1 z) = h(z) + \{h,\phi\}(z) + f_0(z) + \int_0^1 \big\{(1-t)\{h,\phi\} + f_0,\ \phi\big\}(X^t z)\,dt. \tag{1}$$

Source: Pos93 p. 211, displayed formula for $H_0 \circ \Phi$.

Proof. **Step S6.16: Taylor.** Apply the Taylor formula with integral remainder for C^N functions of one real variable (external fact: induction on N with one integration by parts per step, [K:bp:p1:ext-integral]) to $g(t) = u(X^t z)$ at 0, evaluated at 1, using $g^{(j)}(0) = (\mathrm{ad}_\phi^j u)(z)$ and $g^{(N)}(t) = (\mathrm{ad}_\phi^N u)(X^t z)$ (Lemma 6.7). The bound uses $\int_0^1 (1-t)^{N-1}/(N-1)!\,dt = 1/N!$.

Step S6.17: The displayed formula. Apply the case $N = 2$ to h ($\mathrm{ad}^2 h = \{\{h,\phi\},\phi\}$) and the case $N = 1$ to f_0 and add: the remainders are $\int_0^1 (1-t)\{\{h,\phi\},\phi\}(X^t z)\,dt + \int_0^1 \{f_0,\phi\}(X^t z)\,dt$, and brackets are bilinear. □

6.4. **Lemma B.5: composition in the weighted norm.**

Lemma 6.9 (Iterated brackets)**.** *Let $\phi \in \mathcal{O}(r',s')$, $u \in \mathcal{O}(r,s)$ have finite norms, $0 < \rho < r$, $0 < \sigma < s$, $r + \rho \le r'$, $s + \sigma \le s'$, and $\eta = 2|\phi|_{r',s'}/(\rho\sigma)$. Then for every $h \ge 1$,*

$$|\mathrm{ad}_\phi^h u|_{r-\rho,s-\sigma} \le \Big(\frac{2h^2}{e\rho\sigma}\Big)^h \frac{h!}{(2h)!}\,|\phi|_{r',s'}^h |u|_{r,s}, \qquad \textit{hence} \qquad \frac{1}{h!}|\mathrm{ad}_\phi^h u|_{r-\rho,s-\sigma} \le \Big(\frac{e\eta}{4}\Big)^h |u|_{r,s}.$$

Source: Pos93 p. 215 (the displayed estimates for $|\mathrm{ad}_\phi^h u|$ with $\tilde\rho = \rho/h$, $\tilde\sigma = \sigma/h$).

Proof. **Step S6.18: Setup.** Fix $h \ge 1$, $\kappa = \rho/h$, $\lambda = \sigma/h$, $r_i = r - i\kappa$, $s_i = s - i\lambda$ ($0 \le i \le h$), so $r_h = r-\rho$, $s_h = s-\sigma$. By induction each $\mathrm{ad}_\phi^i u \in \mathcal{O}(r_i, s_i)$.

Step S6.19: One step. Let $1 \le i \le h$. Since $\mathrm{ad}_\phi^i u = \{\mathrm{ad}_\phi^{i-1} u, \phi\} = -\{\phi, \mathrm{ad}_\phi^{i-1} u\}$, Lemma 5.7 with $(u, v; r_0, s_0; r, s; \rho, \sigma) \to (\phi, \mathrm{ad}_\phi^{i-1} u; r', s'; r_{i-1}, s_{i-1}; \kappa, \lambda)$ applies (note $\kappa < r_{i-1}$, $r_i < r'$, and likewise for s) and gives $|\mathrm{ad}_\phi^i u|_{r_i,s_i} \le \frac{1}{e}\Big(\frac{1}{(r'-r_{i-1}+\kappa)\lambda} + \frac{1}{(s'-s_{i-1}+\lambda)\kappa}\Big)|\phi|_{r',s'}|\mathrm{ad}_\phi^{i-1} u|_{r_{i-1},s_{i-1}}$. Now $r' - r_{i-1} + \kappa = (r'-r) + i\kappa \ge \rho + i\kappa = (h+i)\kappa$ and likewise $s' - s_{i-1} + \lambda \ge (h+i)\lambda$; so the bracket is $\le \frac{2}{e(h+i)\kappa\lambda}$.

Step S6.20: Product. Multiplying over $i = 1, \dots, h$ and using $\prod_{i=1}^h (h+i) = (2h)!/h!$ and $\kappa\lambda = \rho\sigma/h^2$ gives the first estimate. For the second, $(2h)! \ge (2h/e)^{2h}$ (from $e^N \ge N^N/N!$), so $\frac{1}{h!}|\mathrm{ad}_\phi^h u| \le \frac{(\eta h^2/e)^h}{(2h)!}|u| \le \big(\eta\frac{h^2}{e}\cdot\frac{e^2}{4h^2}\big)^h |u| = (\frac{e\eta}{4})^h |u|$. □

Theorem 6.10 (Lemma B.5: $|u\circ X_\phi^1|\le(1-\eta)^{-1}|u|$)**.** *Let $D\subset\mathbb{C}^n$, $0<\rho<r\le r'-\rho$, $0<\sigma<s\le s'-\sigma$, $\phi\in\mathcal{O}(r',s')$ with finite norm and $\eta=2|\phi|_{r',s'}/(\rho\sigma)\le\frac12$. Put $\Omega=\mathcal{D}_{r'-\rho,s'-\sigma}\supset\mathcal{D}_{r,s}$ and $\nu=\nu_{\rho,\sigma}$.*

(i) $\nu(X_\phi)\le\eta/2\le\frac14$ on Ω; the flow $X^t=X_\phi^t$, $|t|\le1$, of Lemma 6.4 is defined on $\Omega^{1/2}\supset\mathcal{D}_{r-\rho,s-\sigma}$, with $X^t(\mathcal{D}_{r-\rho,s-\sigma})\subset\mathcal{D}_{r,s}$.

(ii) For every $u\in\mathcal{O}(r,s)$ with finite norm, $\sum_{h\ge0}\frac{1}{h!}\operatorname{ad}_\phi^h u$ converges in $|\cdot|_{r-\rho,s-\sigma}$ to a function $S\in\mathcal{O}(r-\rho,s-\sigma)$ and $S=u\circ X_\phi^1$ on $\mathcal{D}_{r-\rho,s-\sigma}$. In particular $u\circ X_\phi^1\in\mathcal{O}(r-\rho,s-\sigma)$ and

$$|u\circ X_\phi^1|_{r-\rho,s-\sigma}\le\Big(1-\frac{e\eta}{4}\Big)^{-1}|u|_{r,s}\le\Big(1-\frac{2}{\rho\sigma}|\phi|_{r',s'}\Big)^{-1}|u|_{r,s}.$$

(iii) The same holds for X_ϕ^t, $0\le t\le1$ (and $-1\le t\le0$), with η replaced by $|t|\eta$.

(iv) If ϕ and u are real on real points, $u\circ X_\phi^1$ is real on real points of $\mathcal{D}_{r-\rho,s-\sigma}$.

Source: Pos93 pp. 214–215, Lemma B.5 (with $|\phi|_{r_0,s_0}<\rho\sigma/2$ there).

Source gap: Pöschel writes "Consider the Lie series expansion $u\circ X_\phi^1=\sum\frac{1}{h!}\operatorname{ad}_\phi^h u$" and estimates the series; the equality with the composition is not proved on p. 214. It is proved here for $\eta\le\frac12$; see Remark 6.1. The remark that the smallness condition implies $X_\phi^t:V_{r-\rho,s-\sigma}\to V_{r,s}$ is (i).

Proof. **Step S6.21: The field and the domain.** Since $r\le r'-\rho$, $s\le s'-\sigma$: $\mathcal{D}_{r,s}\subset\Omega$. Lemma 6.3 (for $\phi\in\mathcal{O}(r',s')$ and losses ρ,σ) gives $\nu(X_\phi)\le|\phi|/(\rho\sigma)=\eta/2=:\varepsilon\le\frac14$ on Ω. If $z\in\mathcal{D}_{r-\rho,s-\sigma}$, then $\bar B_\nu(z,\frac12)=\bar P(z;\sigma/2,\rho/2)\subset\mathcal{D}_{r-\rho/2,s-\sigma/2}\subset\mathcal{D}_{r,s}\subset\Omega$ (Definition 5.1(b)), so $z\in\Omega^{1/2}$. By Lemma 6.4(a), $X^t(z)\in\bar P(z;\varepsilon\sigma,\varepsilon\rho)\subset\mathcal{D}_{r-(1-\varepsilon)\rho,s-(1-\varepsilon)\sigma}\subset\mathcal{D}_{r,s}$. This is (i).

Step S6.22: The series. By Lemma 6.9 (applicable: $r+\rho\le r'$, $s+\sigma\le s'$), $\frac{1}{h!}\operatorname{ad}_\phi^h u\in\mathcal{O}(r-\rho,s-\sigma)$ has norm $\le(e\eta/4)^h|u|_{r,s}$, and $e\eta/4<1$. The sum of norms is $\le(1-e\eta/4)^{-1}|u|_{r,s}$. By Lemma 5.4(a) the partial sums converge uniformly on $\mathcal{D}_{r-\rho,s-\sigma}$, so S is holomorphic ((E2)) and periodic, and termwise integration ((E3)) shows that its Fourier coefficients are the sums of the coefficients; the triangle inequality in $\sum_k|\hat S_k(I)|e^{2\pi|k|_1(s-\sigma)}$ gives $|S|_{r-\rho,s-\sigma}\le(1-e\eta/4)^{-1}|u|_{r,s}$; and $1-e\eta/4\ge1-\eta$.

Step S6.23: The remainder tends to 0. Let $z\in\mathcal{D}_{r-\rho,s-\sigma}$, $t\in[0,1]$, $w=X^t(z)$. By Step S6.21, $w\in\mathcal{D}_{r-\hat\rho,s-\hat\sigma}$ with $\hat\rho=(1-\varepsilon)\rho$, $\hat\sigma=(1-\varepsilon)\sigma$, $0<\hat\rho<r$, $r+\hat\rho\le r'$, and likewise for s. Lemma 6.9 with losses $(\hat\rho,\hat\sigma)$ and $\hat\eta=2|\phi|/(\hat\rho\hat\sigma)=\eta/(1-\varepsilon)^2\le\frac{16}{9}\eta\le\frac89$ gives $|\operatorname{ad}_\phi^N u|_{r-\hat\rho,s-\hat\sigma}\le N!(e\hat\eta/4)^N|u|_{r,s}$, and $e\hat\eta/4\le\frac{2e}{9}<1$. By Lemma 5.4(a), $|(\operatorname{ad}_\phi^N u)(X^tz)|\le N!(e\hat\eta/4)^N|u|_{r,s}$. Lemma 6.8 (with $\Omega_1=\mathcal{D}_{r,s}$, Ω_1-trajectory by Step S6.21) gives $|R_N(z)|\le(e\hat\eta/4)^N|u|_{r,s}\to0$. Hence $u(X^1z)=\lim_N\sum_{j<N}\frac{1}{j!}(\operatorname{ad}_\phi^j u)(z)=S(z)$. This proves (ii).

Step S6.24: (iii), (iv). $X_\phi^t=X_{t\phi}^1$ (the field $X_{t\phi}=tX_\phi$ and $s\mapsto X_\phi^{ts}(z)$ solves it), and $|t\phi|=|t||\phi|$, so (iii) is (ii) for $t\phi$. (iv): Lemma 6.4(e) with $F=X_\phi$ real on real points (a real-valued ϕ on real points has real partials there). □

Proposition 6.11 (Lie transform with first-order remainder, in the weighted norm)**.** *In the setting of Theorem 6.10, let $h\in\mathcal{O}(r,s)$ (any norm, possibly ∞) with $w:=\{h,\phi\}\in\mathcal{O}(r,s)$ of finite norm, and $f_0\in\mathcal{O}(r,s)$ with finite norm. Then on $\mathcal{D}_{r-\rho,s-\sigma}$*

$$(h+f_0)\circ X_\phi^1=h+w+f_0+\mathcal{R},\qquad\mathcal{R}=\sum_{j\ge1}\Big(\frac{1}{(j+1)!}\operatorname{ad}_\phi^j w+\frac{1}{j!}\operatorname{ad}_\phi^j f_0\Big)=\int_0^1\big\{(1-t)w+f_0,\phi\big\}\circ X_\phi^t\,dt,$$

the series converges in $|\cdot|_{r-\rho,s-\sigma}$, $\mathcal{R}\in\mathcal{O}(r-\rho,s-\sigma)$, and

$$|\mathcal{R}|_{r-\rho,s-\sigma}\le\sum_{j\ge1}\Big(\frac{e\eta}{4}\Big)^j\Big(\frac{|w|_{r,s}}{j+1}+|f_0|_{r,s}\Big)\le\frac{\eta}{1-\eta}\Big(\tfrac12|w|_{r,s}+|f_0|_{r,s}\Big).$$

The integral form is Lemma 6.8, (1); the pointwise identity requires no bound on h itself.

Source: Pos93 p. 211 (the decomposition $H_0 \circ \Phi = h + \{h,\phi\} + f_0 + \int \dots$) and p. 212–213 (estimate of f_+); the series form of the remainder is ours.

Proof. **Step S6.25: Derivatives of h.** Since $\mathrm{ad}_\phi^1 h = w$, $\mathrm{ad}_\phi^j h = \mathrm{ad}_\phi^{j-1} w$ ($j \ge 1$). In Lemma 6.7 (with $\Omega_1 = \mathcal{D}_{r,s}$) $g(t) = h(X^t z)$ has $g^{(j)}(t) = (\mathrm{ad}_\phi^{j-1} w)(X^t z)$; Taylor's formula to order N gives $h(X^1 z) = h(z) + \sum_{i=0}^{N-2} \frac{1}{(i+1)!}(\mathrm{ad}_\phi^i w)(z) + R'_N$, with $|R'_N| \le \frac{1}{N!}\max_t |(\mathrm{ad}_\phi^{N-1} w)(X^t z)|$. By Step S6.23 (Lemma 6.9 with $u = w$ and the losses $\hat\rho, \hat\sigma$) this is $\le \frac{(N-1)!}{N!}(e\hat\eta/4)^{N-1}|w|_{r,s} \to 0$. So $h \circ X^1 = h + \sum_{i\ge 0} \frac{1}{(i+1)!}\mathrm{ad}_\phi^i w$ pointwise, the term $i = 0$ being w.

Step S6.26: Derivatives of f_0. By Theorem 6.10(ii), $f_0 \circ X^1 = \sum_{j\ge 0} \frac{1}{j!}\mathrm{ad}_\phi^j f_0$.

Step S6.27: Sum and bound. Adding gives the first identity with $\mathcal{R}$ as stated (terms $j = 0$ are w, f_0). Lemma 6.9 bounds $\frac{1}{j!}|\mathrm{ad}_\phi^j w| \le (e\eta/4)^j |w|$, hence $\frac{1}{(j+1)!}|\mathrm{ad}_\phi^j w| \le \frac{1}{j+1}(e\eta/4)^j|w|$, and similarly for f_0. The geometric sum with $\frac{1}{j+1} \le \frac{1}{2}$ and $e\eta/4 \le \eta$ gives the last bound. Convergence in the norm and membership of $\mathcal{O}(r-\rho, s-\sigma)$ are as in Step S6.22.

Step S6.28: Integral form. Equality with the integral is (1), because $\{h,\phi\} = w$; the trajectory stays in $\mathcal{D}_{r,s}$ by Step S6.21. □

Corollary 6.12 (Integral remainders in the weighted norm)**.** *In the setting of Theorem 6.10 let $v_t \in \mathcal{O}(r,s)$ ($0 \le t \le 1$) have finite norms with $V = \sup_t |v_t|_{r,s} < \infty$, and suppose $(t,z) \mapsto v_t(z)$ and $(t,z) \mapsto Dv_t(z)$ are continuous on $[0,1] \times \mathcal{D}_{r,s}$. Then $G(z) = \int_0^1 v_t(X_\phi^t z)\,dt$ lies in $\mathcal{O}(r-\rho, s-\sigma)$ and $|G|_{r-\rho,s-\sigma} \le (1-\eta)^{-1}V$.*

Source: Pos93 p. 212–213 ("$|u \circ X_\phi^t| \le (1 - 2|\phi|/\rho\sigma)^{-1}|u|$ uniformly for $0 \le t \le 1$", applied inside the integral).

Proof. **Step S6.29: Integrand.** $g_t = v_t \circ X^t \in \mathcal{O}(r-\rho, s-\sigma)$ with $|g_t| \le (1-t\eta)^{-1}V \le (1-\eta)^{-1}V$ (Theorem 6.10(iii)). The map $(t,z) \mapsto g_t(z)$ is continuous and $D_z g_t = Dv_t(X^t z) D_z X^t(z)$ is continuous (Lemma 6.4(b)).

Step S6.30: Holomorphy and coefficients. By (E1), G is holomorphic; it is periodic. Fubini on $[0,1] \times [0,1]^n$ gives $\hat G_k(I) = \int_0^1 (\hat{g_t})_k(I)\,dt$, so by Tonelli $|G|_{I,s-\sigma} \le \int_0^1 |g_t|_{I,s-\sigma}\,dt \le (1-\eta)^{-1}V$. □

6.5. The canonical map $\Phi = X_\phi^1$ on weighted domains.

Proposition 6.13 (Iterative-Lemma package for Φ)**.** *Let $D \subset \mathbb{C}^n$, $\phi \in \mathcal{O}(r,s)$ with finite norm, $0 < \rho < r/2$, $0 < \sigma < s/2$, and $\varepsilon := |\phi|_{r,s}/(\rho\sigma) \le \frac{1}{2}$. Put $\Omega = \mathcal{D}_{r-\rho,s-\sigma}$, $\nu = \nu_{\rho,\sigma}$, $\Phi = X_\phi^1$, $\Psi = X_\phi^{-1}$ (Lemma 6.4). Then:*

- *(a) $\nu(X_\phi) \le \varepsilon$ on Ω; Φ, Ψ are defined and holomorphic on $\mathcal{D}_{r-\frac{3}{2}\rho,s-\frac{3}{2}\sigma} \subset \Omega^{1/2}$ and map it into Ω. For every $t \in [-1,1]$, $\nu(X^t - \mathrm{id}) \le |t|\varepsilon$; explicitly $|\Phi_\theta - \theta|_\infty \le |\phi|_{r,s}/\rho$ and $|\Phi_I - I|_\infty \le |\phi|_{r,s}/(e\sigma)$, and the same for Ψ.*
- *(b) $\Phi(\theta + k, I) = \Phi(\theta, I) + (k, 0)$ for $k \in \mathbb{Z}^n$; if ϕ is real on real points, Φ maps real points to real points.*
- *(c) (Derivative) on $\mathcal{D}_{r-2\rho,s-2\sigma}$: $|D\Phi - \mathbf{1}|_\nu \le 2\varepsilon$, i.e. $|W(D\Phi - \mathbf{1})W^{-1}| \le 2|\phi|_{r,s}/(\rho\sigma)$ for $W = \mathrm{diag}(\sigma^{-1}, \rho^{-1})$.*
- *(d) (Canonical) $(D\Phi)^\top J\, D\Phi = J$, and Corollary 6.6 holds for Φ.*
- *(e) (Inverse) $\Psi \circ \Phi = \mathrm{id} = \Phi \circ \Psi$ on $\mathcal{D}_{r-2\rho,s-2\sigma}$, and $\Phi(\mathcal{D}_{r-2\rho,s-2\sigma}) \subset \mathcal{D}_{r-\frac{3}{2}\rho,s-\frac{3}{2}\sigma}$.*

Source: Pos93 p. 212 ("Equivalently, X_ϕ satisfies $|WX_\phi|_{\mathcal{P}} \le \frac{q\delta}{\alpha\rho\sigma} \le \frac{1}{2}$ … X_ϕ^t maps $V_{r-\frac{3}{2}\rho,s-\frac{3}{2}\sigma}$ into $V_{r-\rho,s-\sigma}$ … $|W(DX_\phi^t - I_{2n})W^{-1}| \le 2\frac{q\delta}{\alpha\rho\sigma}$ uniformly on $V_{r-2\rho,s-2\sigma}$"), with $|\phi|_{r,s}$ in place of $q\delta/\alpha$.

Proof. **Step S6.31: Field.** Lemma 6.3 gives $\nu(X_\phi) \le \varepsilon \le \frac{1}{2}$ on Ω, and Ω is open. For $z \in \mathcal{D}_{r-c\rho,s-c\sigma}$ ($c \ge \frac{3}{2}$) the ball $\bar{B}_\nu(z, c-1) = \bar{P}(z; (c-1)\sigma, (c-1)\rho)$ lies in Ω (Definition 5.1(b)): so $\mathcal{D}_{r-\frac{3}{2}\rho,s-\frac{3}{2}\sigma} \subset \Omega^{1/2}$ and $\mathcal{D}_{r-2\rho,s-2\sigma} \subset \Omega^1$.

Step S6.32: (a),(b). Lemma 6.4(a),(b),(d),(e) (with $\tau = (k, 0)$: $\Omega + \tau = \Omega$ and X_ϕ is periodic, as ϕ is). $X^t(z) \in \bar{P}(z; \varepsilon\sigma, \varepsilon\rho) \subset \mathcal{D}_{r-\rho,s-\sigma}$ for $z \in \mathcal{D}_{r-\frac{3}{2}\rho,s-\frac{3}{2}\sigma}$ because $\varepsilon \le \frac{1}{2}$. The explicit component bounds are $|X^t_\theta - \theta| \le |t| \sup|\partial_I \phi| \le |t||\phi|/\rho$ and $|X^t_I - I| \le |t| \sup|\partial_\theta \phi| \le |t||\phi|/(e\sigma)$ (mean value inequality along the trajectory and Lemma 6.3). For $\Psi = X_\phi^{-1} = X^1_{-\phi}$ the same with $-\phi$.

Step S6.33: (c)–(e). (c) is Lemma 6.4(c) with Ω^1 and Step S6.31. (d) is Lemma 6.5 and Corollary 6.6. (e): Lemma 6.4(c) for $z \in \Omega^1$ gives $\Psi(\Phi z) = z$ and $\Phi(\Psi z) = z$; and $\Phi(z) \in \bar{B}_\nu(z, \varepsilon) \subset \mathcal{D}_{r-\frac{3}{2}\rho,s-\frac{3}{2}\sigma}$ for $z \in \mathcal{D}_{r-2\rho,s-2\sigma}$ ($\varepsilon \le \frac{1}{2}$). □

Remark 6.14 (Use by Part 6 and track P; what is not provided). Part 6 applies Proposition 6.11 with h the integrable part, $w = \{h, \phi\} = g_+ - f_0$ by the homological equation (so $|w|$ is controlled by $|\chi|$ and $|T_K f|$, not by $|h|$), $f_0 = g + T_K f$, and Theorem 6.10 for $(f - T_K f) \circ X^1_\phi$, with the losses (ρ, σ) from (r, s) to $(r-\rho, s-\sigma)$ (so ϕ lives on $(r+\rho, s+\sigma)$, or one re-indexes as Pöschel: $\phi \in \mathcal{O}(r, s)$, $u \in \mathcal{O}(r-\rho, s-\sigma)$, result on $\mathcal{D}_{r-2\rho,s-2\sigma}$). The smallness used is $\eta = 2|\phi|/(\rho\sigma) \le \frac{1}{2}$; the displacement and derivative bounds of Proposition 6.13 need only $\varepsilon \le \frac{1}{2}$. The sup-norm *fallback* of `PLAN.md` is not used. Not proved here: the stability transfer (Part 7) and the estimate of $|W(\Phi_1 \circ \cdots \circ \Phi_N - \mathrm{id})|$ for compositions of the maps of successive steps, which is Part 6's (the displacement bounds of (a) add, since each Φ_i maps into the domain of Φ_{i+1} by the choice of losses).